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REGULAR PAPER

Time-delayed dual-rate haptic rendering: stability analysis and reduced order modeling

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Abstract
This work presents an exhaustive analysis of the impact of time delay on the stability of dual-rate haptics controllers using an exact discrete-time method. The mathematical formulation of such controllers leads to higher order state-space models, in particular, for higher values of time delay and sampling rates. A balanced truncation based model order reduction framework is therefore utilized for obtaining reduced order models, while preserving the stability properties of the original higher order models. The likely order of the reduced models is selected on the basis of the Hankel singular values which represent the contribution of the system states towards the overall system energy. Using this framework, it is empirically found that third order reduced models yield exactly the same stability ranges as given by the allied full order models. This empirical finding is backed up by a rigorous analysis of the controller while considering a wide range of values for the time delay spanning both the realistic and worst-case application scenarios. This result is hitherto unknown in the haptics literature and is for the first time reported in this paper. For comparison purposes, the stability ranges of the dual-rate controller are also obtained using an equivalent continuous-time method, and numerical simulations. This work generalizes the results of previous works available in the literature for uniform-rate sampling scheme both for delayed and non-delayed haptics controllers. The results demonstrate that for a time-delayed dual-rate haptics controller, an increase in the sampling rate leads to an enhancement in the stable range of virtual wall parameters as long as the value of time delay relative to the sampling rate (non-dimensional time delay) remains small. For higher values of the non-dimensional time delay, higher sampling rates do not necessarily lead to performance enhancement.

Keywords Dual-rate haptics controllers · Time delay · State-space modeling · Model order reduction · Balanced truncation · Hankel singular values · Impedance haptic interfaces

1 Introduction

Haptic interfaces enable a human operator to simultaneously perceive and to act upon a virtual (or distant) environment using the sense of touch. These interfaces make physical interactions with the virtual objects feel more natural and realistic and have thereby found widespread applications in a multitude of domains, for instance in the medical field (Wang et al. 2017), education (Okamura et al. 2002), training (Abate et al. 2009), entertainment (Mazzoni and Bryan-Kinns 2016), virtual prototyping (Ix et al. 2001), forensics (Buck et al. 2008) etc.

A haptic interface typically provides kinesthetic feedback to a user wherein a force proportional to the impedance of a virtual environment is applied, in response to an admissible motion input. It is well known (Colgate and Brown 1994; Diolaiti et al. 2006), that attempts made to emulate stiff virtual environments often lead to device instability, yet rendering high stiffness is desirable because of human perception thresholds (Tan et al. 1994), and practical considerations (Wang et al. 2011). Factors that are responsible for this inadequacy of haptic devices have been identified in Minsky et al. (1990); Colgate and Brown (1994), Diolaiti

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et al. (2006), Gil et al. (2009), Díaz and Gil (2010), Hulin et al. (2014), Chawda et al. (2015). These works have highlighted that the most dominant factors that impede the performance of haptic interfaces are the controller sampling rate, time delay, device inherent damping and compliance.

Several strategies have been proposed that enable uniform-rate haptics controllers to render high stiffness (Vasudevan et al. 2007; Weir et al. 2008; Chawda et al. 2011; Ryu and Yoon 2014). The more recent efforts being made in Song et al. (2016) using the co-actuation strategy, in Desai et al. (2019) using model matching framework, in Gil and Díaz (2019) using voltage-mode control, and in Singh et al. (2019) by utilizing the successive force augmentation approach. An alternative approach to enhance performance, particularly at higher sampling rates, and for deformable environment rendering, relies on the use of multi-rate controllers (Lee and Lee 2004; Koul et al. 2017; Fotoohi et al. 2007; Cavusoglu and Tendick 2000; Barbagli et al. 2002). Multi-rate control strategies offer several advantages compared to their uniform-rate counterparts. Most importantly, such strategies do not demand extra hardware as is the case in An and Kwon (2006), Weir et al. (2008), Gosline and Hayward (2008), or device design alterations as in Song et al. (2016) or any complex algorithm as in Chawda et al. (2011), besides they can be easily incorporated with the existing hardware setups. Among the multi-rate control strategies mentioned above, here, we are primarily interested in the dual-rate control scheme proposed in Koul et al. (2017). In the dual-rate haptics controller, a virtual wall is rendered at two distinct sampling rates unlike the more generic case in which a uniform sample rate is used. Specifically, the stiffness of the virtual wall is rendered at higher sampling rates, while, the damping is rendered at relatively lower sample rates. With such a decoupled rendering scheme, higher wall stiffnesses have been emulated than those admitted by the uniform-rate scheme (Koul et al. 2017). The motivation for using such a control scheme is to avert the performance degradation that haptics controllers experience at high sampling rates owing to poor velocity estimation from digital position sensors, especially the ones with poor resolutions. Several non-conventional velocity estimation techniques have also been proposed to tackle the same problem, however, such methods may not be useful at very higher sampling rates (for e.g., > 10 kHz), and may also necessitate fine tuning of the involved parameters. For instance, Hayward et al. (1997) proposed an adaptive windowing technique that adaptively changes the window length of position values used for the determination of velocity. The algorithm adapts in response to the magnitude of input velocity and optimizes the tradeoff between delay, transient response and noise reduction. Zhou et al. (2008), however, argue that at higher sampling rates the adaptive windowing technique may be biased in selecting certain window lengths preferentially over others and consequently yield poor velocity estimates. A similar observation is made by Sinclair et al. (2014) during stick-slip rendering in coupled audio-haptic simulations. Due to the stick-slip phenomenon, the adaptive windowing was expected to have mostly assumed small window lengths, however, for nearly 98% of time steps the algorithm used the maximum window length and thus induced time delay in the loop. Chawda et al. (2011) proposed the use of a sliding mode control based Levant's differentiator for velocity estimation for a 1-DoF haptic interface. Similarly, Ghaffari and Kövecses (2013) proposed the use of curve breaking velocity estimation (CBVE) algorithm that aims to minimize noise and time delay in velocity estimates owing to discretization and quantization noise. In Chawda et al. (2018) a comparative evaluation of various velocity estimation methods including the finite difference method (FDM), Levant's differentiator and adaptive windowing is reported. Their results suggest that the use of Levant's differentiator results in the rendering of highest stiffness, FDM follows next, while the adaptive windowing yields the minimum rendered stiffness. Furthermore, FDM enables the rendering of maximum virtual damping while Levant's differentiator and adaptive windowing yield the minimal virtual damping. Adaptive windowing however outperforms FDM and Levant's differentiator with regard to fidelity error and it has no tunable parameters. In all these non-conventional velocity estimation strategies relatively smaller sample rates have been used as compared to the dual-rate control scheme that has proven effective for sample rates as high as 20 kHz (Koul et al. 2017). Besides this, a distinguishing feature of the dual-rate control scheme as compared to these alternate methods is that it allows for the independent implementation of the virtual stiffness at elevated sample rates. For the majority of applications of haptic devices, that is the most desirable property the users wish to emulate.

Digital realization of haptic rendering incurs inevitable time delays, primarily attributed to the spatio-temporal discretization due to sampling and quantization (Gillespie and Cutkosky 1996; Gil et al. 2009; Çavuşoğlu et al. 2002; Martin and Hillier Citeseer (2009).). Data communication, virtual contact estimation, velocity filtering and amplifier dynamics are some other sources that introduce additional delays into the haptic loop (Díaz and Gil 2010; Diolaiti et al. 2006; Chawda et al. 2015). For instance, a delay of about 2–5 sample periods was found in a commercial haptic device Novint Falcon (Martin and Hillier Citeseer (2009).), and for Phantom haptic device a delay of about 50 sample periods has been reported (Çavuşoğlu et al. 2002). Time delay is known to aggravate the energy instilling tendency (Gillespie and Cutkosky 1996) of the digital implementation and thus has been recognized as a major impediment affecting the stability and rendering quality of haptic interaction. Prior research has shown that delay causes instability (Salcudean and Vlaar 1994; Diolaiti et al. 2006; Hulin et al. 2014; Colonnese and Okamura 2016), reduces transparency (Hirche et al. 2005), and even distorts the perception of the simulated environment properties (Pressman et al. 2007; Nisky et al. 2008). While the effect of time-delay on the performance of uniform-rate haptics controllers has been well understood, however, little is known about its effect on the performance of multi-rate haptics controllers. In particular, as per the author's knowledge, no work has yet been attempted to determine the effect of time delay on the performance of dual-rate haptics controllers.

Stability of haptic devices is indispensable as haptic interaction by its very fundamental nature involves the presence of a human operator in the control loop, be it during a virtual interaction or in a master-slave teleoperation setting. A knowledge of the stability bounds of a haptics controller is thus essential. Stability analysis of haptics controllers has been studied extensively and is still a prime focus of haptics research, as is evident by the recent works reported in Kim and Lee (2018), Liu et al. (2018), Mashayekhi et al. (2018), Mashayekhi et al. (2019). Previous works (Love and Book 1995; Adams and Hannaford 1999; Gil et al. 2004; Díaz and Gil 2010; Tokatli and Patoglu 2015; Gil et al. 2018; Kim and Lee 2018) have analyzed the stability of uniform-rate haptics controllers without accounting for time delay, whereas in works (Salcudean and Vlaar 1994; Diolaiti et al. 2006; Gil et al. 2009; Dang et al. 2016; Liu et al. 2018; Mashayekhi et al. 2018, 2019) time delay has been taken into account. Diolaiti et al. (2006) have obtained conditions for stability of time-delayed haptic rendering subjected to the existence of a maximum initial velocity and acceleration. Stability conditions for time-delayed haptic rendering have been obtained by Gil et al. (2009) for small ranges of device damping, virtual damping and time delay. Mashayekhi et al. (2018) have, however, developed the closed-form equations without any limitation on these parameter ranges. Salcudean and Vlaar (1994) have argued that high stiffness rendering is mostly limited by time delay in lightweight and direct drive haptic systems. Dang et al. (2016) have studied the stability of haptics controllers in presence of time varying delays. More recently, Liu et al. (2018) have analyzed the stability of haptic systems for an uncertain time delay, and Mashayekhi et al. (2019) have carried the delay-dependent stability analysis of haptic rendering using modal transform and free weighting matrices approach. In contrast to this, very few works (Lee and Lee 2004; Dai et al. 2008; Koul et al. 2017), have examined the stability of multi-rate haptics controllers. These works have, however, ignored the time delay in their analysis. We hypothesize that owing to the difference in the basic control architecture of multi-rate and uniform-rate haptics controllers, time delay will have a different effect on the stability of the former than it has on the latter.

In this work, we present the stability analysis of a time-delayed dual-rate haptics controller. In particular, we seek to analyze how the combined presence of time delay and two distinct sampling rates affect the stability boundaries of a dual-rate haptics controller. Based on a linear and mixed discrete-continuous model, we analyze the given controller using the state-space framework and state augmentation approaches. The problem is formulated in terms of discrete-time state-space equations and later Lyapunov theory is used to determine the stability boundaries.

As would become apparent in later sections, the description of such systems leads to higher order state-space models which make their stability analysis complicated, especially for higher loop delays and sampling rates. We therefore employ a model order reduction technique based on balanced truncation to develop lower order models that approximate the behavior of the original higher order models while preserving their stability. It turns out that third order reduced models yield the same stability bounds as predicted by the original full order models, irrespective of the magnitude of time delay and sampling frequency combinations that are considered here. This finding is valid for uniform-rate and dual-rate haptics controllers alike. It is pertinent to mention that simple models are very desirable in haptic rendering for online stability evaluation. This is particularly true for applications where the simulated environment parameters are themselves estimated online and are not known beforehand. For instance, in a haptic feedback enabled remote surgery, the rendered stiffness is variable and depends on the contact conditions i.e., whether the user is interacting with a muscle, a normal tissue, an abnormal tissue (tumor), an organ, a bone, or another instrument. In such scenarios, the environmental stiffness is usually estimated online first (Zarrad et al. 2007) before attempting to emulate that via a haptic device. Once the environmental stiffness is estimated online, numerical stability evaluation can reveal whether that stiffness can be rendered by the device or not. Similarly, simulating a drilling procedure where the environmental stiffness is highly variable and unknown needs an online stiffness estimation (Shah and Polushin 2013). The stability analysis is also presented using an equivalent continuous-time method, whose validity is restricted to low input frequency, an assumption that can be safely made for haptic rendering of moderate stiffness. The results obtained using this method follow the same trend as observed in the exact discrete-time method, however, its results are less conservative.

2 System architecture and assumptions

Haptics controllers are typical non-linear sampled-data systems as they contain non-linear phenomenon like sensor quantization, Coulomb friction and unilateral constraint as part of the loop, besides containing elements that are mixed continuous and discrete. This work, however, considers a linear model of the dual-rate haptics controller as earlier works have shown that these non-linearities can be safely avoided from the stability analysis (Diolaiti et al. 2006; Mashayekhi et al. 2018). Figure 1 depicts the basic architecture of such a dual-rate haptics controller. We model the haptic device as a rotational one degree of freedom (DoF) system possessing a lumped mass moment of inertia I and viscous damping b. The continuous-time transfer function of the device with net torque as input and rotational displacement as output is accordingly given by, $D(s) = \frac{1}{Is^2+bs}$. We assume that an active actuator powers the device and does not saturate for the rendered feedback gains. The rotational displacement of the device θ is measured by an integrated position sensor while the velocity is determined using backward difference on two successive position values. The virtual wall is modeled as a simple spring-damper system with stiffness coefficient K and damping coefficient B.

The controller samples the position data with a sample period T_K (sampling frequency f_K) and uses this sampled position for rendering the virtual stiffness K via the position loop (inner feedback loop), while as, for implementing the virtual damping B, the position data is sampled with a sample period T_B (sampling frequency f_B) via the velocity loop (outer feedback loop). For mathematical convenience, we assume the sample periods to be integrally related, i.e., $T_B = NT_K$ (equivalently $f_K = Nf_B$), with $N \in \mathbb{Z}^+$. Owing to the high sampling rates employed in the dual-rate haptics controllers (Koul et al. 2017), we assume that no inter-sample information is lost. Furthermore, due to the low-pass filtering characteristics of the haptic device dynamics (Adams and Hannaford 1999), and the fact that force feedback bandwidth is relatively smaller than the typical sampling rates employed (Ryu et al. 2008), the aliasing effects in the dual-rate haptics controller can also be assumed to be insignificant. The effects of aliasing are however important in haptic rendering scenarios that employ either lower sampling rates, for instance, in synchronized haptic-visual rendering (Mahvash and Hayward 2004), or in tactile haptic devices that render high frequency vibration based feedback (Kim et al. 2015). The feedback torques τ_K and τ_B are converted into continuous-time signals using the zero-order hold (*ZOH*) blocks i.e., they remain constant for the duration of their respective sample periods. The torque exerted by the human operator is represented by τ_H.

Owing to the linearity of the system, the time delays in the feedback loops have been lumped together and are represented by their discrete-time transfer functions $z_1^{-\lambda_K}$ and $z_2^{-\lambda_B}$ (Gil et al. 2009; Mashayekhi et al. 2019). The discrete-time variables z_1 and z_2 correspond to sample periods T_K and T_B, respectively. The terms λ_K and λ_B represent the number of sample periods by which the feedback torques τ_K and τ_B are delayed. For a constant time delay of λ in the feedback loops, it follows that $\lambda = \lambda_K T_K = \lambda_B T_B$, which gives $\lambda_K = N\lambda_B$. For simplicity, we further assume that $[\lambda_K, \lambda_B] \in \mathbb{Z}^+$.

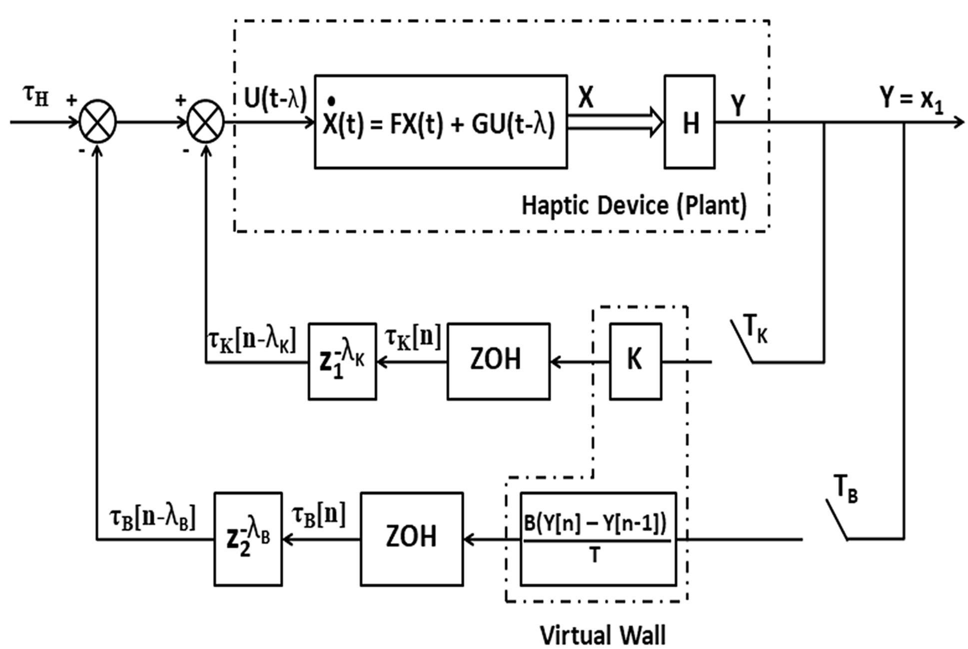


Fig. 1 A block diagram depicting the basic control architecture of dual-rate haptics controller. This linear model takes into account the device dynamics, sampling, zero-order holders and time delays. The inner and the outer feedback loops implement the virtual stiffness K, and virtual damping B, at sample periods of T_K and T_B, respectively

3 Stability analysis using exact discrete-time method

In this section, the stability of the dual-rate haptics controller depicted in Fig. 1 is analyzed using an exact discrete-time method wherein the controller is represented as a pure discrete-time system. The discrete-time system exhibits similar response as the original hybrid controller at the sampling instants (Franklin et al. 1998).

The controller is mathematically represented in terms of discrete-time state-space equations, which are subsequently used for obtaining its stability boundaries. Due to the sampled-data nature of the controller, presence of time delay terms and two distinct sample rates of the feedback loops, the state-space formulation requires transformations like continuous to discrete (discretization), discrete to discrete (resampling), besides requiring the appropriate augmentation of the state vector. Once the problem is formulated into standard state-space representation, discrete-time Lyapunov theory is used to determine the stability limits of the controller.

The differential equations that govern the behavior of the plant while implementing a virtual wall using time-delayed dual-rate haptics controller are given as:

$$\begin{cases} I\ddot{\theta}(t) + b\dot{\theta}(t) = \tau_H(t) - \tau_K(t-\lambda) - \tau_B(t-\lambda) : & \textit{Inside the wall} \\ I\ddot{\theta}(t) + b\dot{\theta}(t) = \tau_H(t) : & \textit{Outside the wall} \end{cases} \tag{1}$$

where, $\ddot{\theta}, \dot{\theta}$ and λ denote the angular acceleration, angular velocity of the haptic interface, and the magnitude of time delay in the controller. The magnitude of torque is positive if it tends to push the device into the wall and negative otherwise. Furthermore, as mentioned in the previous section, the time delay λ is related to λ_K, λ_B, T_K and T_B as:

$$\lambda = \lambda_K T_K = \lambda_B T_B \tag{2}$$

Following the *ZOH* assumption, the feedback torques τ_K and τ_B are governed by:

$$\tau_K(t) = \tau_K(kT_K) = K\theta[kT_K] \quad \forall\, t \in [kT_K, (k+1)T_K) \tag{3}$$

$$\tau_B(t) = \tau_B(kT_B) = B\left\{ \frac{\theta[kT_B] - \theta[(k-1)T_B]}{T_B} \right\} \quad \forall\, t \in [kT_B, (k+1)T_B) \tag{4}$$

where, $k \in \mathbb{Z}^+$ signifies the kth sampling instant. Furthermore, the term $\left\{ \frac{\theta[kT_B] - \theta[(k-1)T_B]}{T_B} \right\}$ represents the device velocity in accordance with the backward difference method.

A haptic device tends to be unstable (for certain parameter combinations) when a sustained contact is made with the virtual wall. Hence, for the purpose of stability analysis, we need to consider only the differential equation that governs its dynamics *inside the wall*. In order to represent the said differential equation in state-space form, we define two state variables as follows:

$$x_1 := \theta \quad \textit{and} \quad x_2 := \dot{\theta} \tag{5}$$

With this choice of state variables, the continuous-time state-space equations of the plant can be written as:

$$\begin{cases} \dot{\mathbf{X}}(t) = \mathbf{F}\mathbf{X}(t) + \mathbf{G}U(t-\lambda) \\ Y(t) = \mathbf{H}\mathbf{X}(t) \end{cases} \tag{6}$$

where, $\mathbf{X} \in \mathbb{R}^{2\times 1}$ is the state vector, $U \in \mathbb{R}$ is the control input and $Y \in \mathbb{R}$ is the system output. Furthermore, $\mathbf{F} \in \mathbb{R}^{2\times 2}$ represents the state matrix, $\mathbf{G} \in \mathbb{R}^{2\times 1}$ is the control vector and $\mathbf{H} \in \mathbb{R}^{1\times 2}$ represents the output vector. Here, they are given by:

$$\mathbf{X} = \left\{x_1 \ x_2\right\}^T; \mathbf{F} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{b}{I} \end{bmatrix}; \mathbf{G} = \left\{0 \ \frac{1}{I}\right\}^T \quad \text{and} \quad \mathbf{H} = \left\{1 \ 0\right\} \tag{7}$$

The instability in haptic devices is mostly observed at frequencies that are outside the bandwidth of human voluntary movement, and that of the involuntary tremor (Shimoga 1992). In fact, past research has shown that a human operator can, in principle, increase the stability of the haptic device by imposing his/her own hand impedance onto the device dynamics (Hulin et al. 2008; Gil et al. 2004). In what follows, we shall thus consider the worst case scenario by assuming that the human operator holds the device with no or minimal grasping force/torque i.e., $\tau_h = 0$. It is to be noted, however, that a human operator can behave actively at low manipulation frequencies [below 10 Hz (Diolaiti et al. 2006)] and exhibit the tendency to destabilize the haptic interaction for certain arm postures, grasping types, and for extended durations of interaction (Dyck et al. 2013; Miandashti and Tavakoli 2014). In such scenarios, the inclusion of a human grasping model in the stability analysis becomes necessary.

Noting that the controller involves two sampling rates, we first need to discretize the plant in Eq. (6), as per the sampling period of the position loop, T_K, and then close the inner feedback loop i.e., the position loop. Afterwards, the resulting system has to be resampled to sample period T_B, and subsequently, the outer feedback loop i.e., the velocity loop has to be closed. Moreover, in order to take care of the time delay terms $z_1^{-\lambda_K}$ and $z_2^{-\lambda_B}$ in the position and velocity feedback loops, the resulting state vectors have to be augmented appropriately.

Now, as the control inputs to the plant are piecewise constant and are preceded by *ZOH* blocks, the discretization of Eq. (6), with respect to the sample period T_K, leads to:

$$\begin{cases} \mathbf{X}[(n+1)T_K] = \mathbf{\Phi}\mathbf{X}[nT_K] + \mathbf{\Gamma}\left(-\tau_K[(n-\lambda_K)T_K]\right) + \mathbf{\Gamma}\left(-\tau_B\left[(n-\lambda_B)T_K\right]\right) \\ \quad Y[nT_k] = \mathbf{\Psi}\mathbf{X}[nT_k] \end{cases} \tag{8}$$

where, $n \in \mathbb{Z}^+$ represents the *nth* sampling instant with respect to the sample period T_K. The same can be represented by $\frac{n}{N}$ with respect to the sample period T_B. The discrete-time representation given by Eq. (8) is causal and signifies the fact that the current state depends on the previous state. It is to be noted that the choice of using $\mathbf{X}[(n+1)T_K]$ and $\mathbf{X}[nT_K]$ for the current and previous state is just for brevity, the states could well be written as $\mathbf{X}[nT_K]$ and $\mathbf{X}[(n-1)T_K]$, respectively. The discretized matrices $\mathbf{\Phi}$, $\mathbf{\Gamma}$ and $\mathbf{\Lambda}$ in Eq. (8), are given by Franklin et al. (1998):

$$\mathbf{\Phi} = e^{\mathbf{F}T_K}; \mathbf{\Gamma} = \left(\int_0^{T_K} e^{\mathbf{F}\eta} d\eta\right)\mathbf{G} \quad \text{and} \quad \mathbf{\Psi} = \mathbf{H} \tag{9}$$

The expression for the discretized matrix $\mathbf{\Phi}$ involves the exponential of its continuous-time counterpart $\mathbf{F}$, which is given by:

$$\mathbf{\Phi} = \sum_{j=0}^{\infty} \frac{1}{j!}\mathbf{F}^j T_K^j \tag{10}$$

For the state matrix $\mathbf{F}$ (given by Eq. (1)), few exponential computations reveal that:

$$\mathbf{F^j} = \begin{bmatrix} 0 & (-\frac{b}{I})^{j-1} \\ 0 & (-\frac{b}{I})^j \end{bmatrix} \quad : for\ j > 0 \tag{11}$$

Accordingly, the discretized version of $\mathbf{F}$, in Eq. (9), i.e., $\mathbf{\Phi}$ is given by:

$$\mathbf{\Phi} = \begin{bmatrix} 1 & T_K - \frac{bT_K^2}{2!I} + \frac{b^2T_K^3}{3!I^2} - \frac{b^3T_K^4}{4!I^3} + \cdots \\ 0 & 1 - \frac{bT_K}{I} + \frac{b^2T_K^2}{2!I^2} - \frac{b^3T_K^3}{3!I^3} + \frac{b^4T_K^4}{4!I^4} + \cdots \end{bmatrix} \tag{12}$$

Using Maclaurin series expansion, the above matrix can be simplified to:

$$\mathbf{\Phi} = \begin{bmatrix} 1 & \frac{I\left(1-e^{-\frac{bT_K}{I}}\right)}{b} \\ 0 & e^{-\frac{bT_K}{I}} \end{bmatrix} \tag{13}$$

Accordingly, the discretized matrix $\mathbf{\Gamma}$, in Eq. (9), can be expressed as:

$$\mathbf{\Gamma} = \left(\int_0^{T_K} \begin{bmatrix} 1 & \frac{I\left(1-e^{-\frac{b\eta}{I}}\right)}{b} \\ 0 & e^{-\frac{b\eta}{I}} \end{bmatrix} d\eta\right) \begin{Bmatrix} 0 \\ \frac{1}{I} \end{Bmatrix} \tag{14}$$

which after simplification leads to:

$$\mathbf{\Gamma} = \begin{bmatrix} \frac{T_K}{b} + \frac{I}{b^2}\left(e^{\frac{-bT_K}{I}} - 1\right) \\ \frac{1}{b}\left(1 - e^{\frac{-bT_K}{I}}\right) \end{bmatrix} \tag{15}$$

Recalling, that the feedback torque due to virtual stiffness is given as (see Eq. (3)):

$$\begin{aligned} \tau_K[(n-\lambda_K)T_K] &= KY[(n-\lambda_K)T_K] \\ &= K\mathbf{\Psi}\mathbf{X}[(n-\lambda_K)T_K] \end{aligned}$$

The discrete-time state space equations, given by Eq. (8), can therefore be written as:

$$\begin{cases} \mathbf{X}[(n+1)T_K] = \mathbf{\Phi}\mathbf{X}[nT_K] - \mathbf{\Gamma}K\mathbf{\Psi}\mathbf{X}[(n-\lambda_K)T_K] + \mathbf{\Gamma}\left(-\tau_B\left[(n-\lambda_B)T_K\right]\right) \\ \quad Y[nT_K] = \mathbf{\Psi}\mathbf{X}[nT_K] \end{cases} \tag{16}$$

The presence of the delayed feedback in Eq. (16) does not yet permit us to establish the closed loop state-space representation of the position feedback loop. In order to do the same, we augment these state equations by introducing λ_K new state variables, which we define as:

$$\begin{aligned} &x_3[nT_K] := X[(n-\lambda_K)T_K]; x_4[nT_K] := X[(n-\lambda_K+1)T_K]; \\ &x_5[nT_K] := X[(n-\lambda_K+2)T_k]; \\ &\ldots x_{\lambda_K+2}[nT_K] := X[(n-1)T_K] \end{aligned} \tag{17}$$

Based on these definitions it also follows that:

$$\begin{aligned} x_3[(n+1)T_K] &= x_4[nT_K]; x_4[n+1)T_K] \\ &= x_5[nT_K]; \ldots x_{\lambda_K+2}[n+1)T_K] \\ &= X[nT_K] \end{aligned} \tag{18}$$

In accordance with the above definitions, the augmented state equations for the system defined by Eq. (16), are given by:

$$\begin{Bmatrix} \mathbf{X}[(n+1)T_K] \\ x_3[(n+1)T_K] \\ x_4[(n+1)T_K] \\ \cdot \\ \cdot \\ \cdot \\ x_{\lambda_K+1}[(n+1)T_K] \\ x_{\lambda_K+2}[(n+1)T_K] \end{Bmatrix} = \begin{bmatrix} \mathbf{\Phi} & -\mathbf{\Gamma} K \mathbf{\Psi} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 1 & 0 & \dots & 0 \\ \mathbf{0} & \mathbf{0} & 0 & 1 & \dots & 0 \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \mathbf{0} & \mathbf{0} & 0 & 0 & \dots & 1 \\ \mathbf{1} & \mathbf{0} & 0 & 0 & \dots & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{X}[nT_K] \\ x_3[nT_K] \\ x_4[nT_K] \\ \cdot \\ \cdot \\ \cdot \\ x_{\lambda_K+1}[nT_K] \\ x_{\lambda_K+2}[nT_K] \end{Bmatrix} + \begin{Bmatrix} \mathbf{\Gamma} \\ 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \\ 0 \end{Bmatrix} \tau_B\left[\left(n-\lambda_B\right)T_K\right] \tag{19}$$

In a compact form, we can write:

$$\begin{cases} \mathbf{X}_{\mathbf{aug}}[(n+1)T_K] = \mathbf{\Phi}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_K] + \mathbf{\Gamma}_{\mathbf{aug}}\tau_B\left[\left(n-\lambda_B\right)T_K\right] \\ Y[nT_K] = \mathbf{\Psi}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_K] \end{cases} \tag{20}$$

where, $\mathbf{\Psi}_{\mathbf{aug}} = [\mathbf{\Psi}\ 0\ 0\ \dots\ 0]_{1\times(\lambda_K+2)}$. Equation (20) represents the closed-loop dynamics of the position feedback loop. This equation can be used for obtaining the stability bounds for a simple virtual spring rendering wherein the virtual damping coefficient and consequently the damping torque is zero.

$$\mathbf{\Phi}^{\mathbf{r}}_{\mathbf{aug}} = [\mathbf{\Phi}_{\mathbf{aug}}]^N \quad \text{and} \quad \mathbf{\Gamma}^{\mathbf{r}}_{\mathbf{aug}} = \sum_{i=0}^{N-1} [\mathbf{\Phi}_{\mathbf{aug}}]^i \mathbf{\Gamma}_{\mathbf{aug}} \tag{22}$$

Now that the state-space equations have been expressed in terms of a single sample period T_B, the velocity feedback loop can be closed. Recalling, that the feedback torque due to virtual damping is given by:

$$\begin{aligned} \tau_B[(n-\lambda_B)T_B] &= B\left\{\frac{\theta[(n-\lambda_B)T_B] - \theta[(n-\lambda_B-1)T_B]}{T_B}\right\} \\ &= \frac{B}{T_B} H\big(X[(n-\lambda_B)T_B] - X[(n-\lambda_B-1)T_B]\big) \end{aligned}$$

With the above substitution, the state-space equations in Eq. (21) can be rewritten as:

$$\begin{cases} \mathbf{X}_{\mathbf{aug}}[(n+1)T_B] = \mathbf{\Phi}^{\mathbf{r}}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_B] - \mathbf{\Gamma}^{\mathbf{r}}_{\mathbf{aug}}\dfrac{B}{T_B} H\big(X[(n-\lambda_B)T_B] - X[(n-\lambda_B-1)T_B]\big) \\ Y[nT_B] = \mathbf{\Psi}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_B] \end{cases} \tag{23}$$

Now, before proceeding to close the velocity feedback loop, we note that the sample period of Eq. (20) is T_K, while the damping torque is actually updated at a sample period of T_B. This necessitates the discrete to discrete transformation i.e., resampling of Eq. (20) from sample period T_K to sample period T_B. As the sample periods T_K and T_B are integrally related, the resampled state-space equations are given as:

$$\begin{cases} \mathbf{X}_{\mathbf{aug}}[(n+1)T_B] = \mathbf{\Phi}^{\mathbf{r}}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_B] + \mathbf{\Gamma}^{\mathbf{r}}_{\mathbf{aug}}\tau_B\left[\left(n-\lambda_B\right)T_B\right] \\ Y[nT_B] = \mathbf{\Psi}_{\mathbf{aug}}\mathbf{X}_{\mathbf{aug}}[nT_B] \end{cases} \tag{21}$$

where, the resampled matrices $\mathbf{\Phi}^{\mathbf{r}}_{\mathbf{aug}}$ and $\mathbf{\Gamma}^{\mathbf{r}}_{\mathbf{aug}}$ are given as Fotoohi et al. (2007):

As was done earlier, these state-space equations need to be augmented in order to take care of the delay in the feedback loop. The same is accomplished by using new $\lambda_B + 1$ state variables, which are defined as:

$$\begin{aligned} &x_{\lambda_K+3}[nT_B] := X[(n-\lambda_B-1)T_B]; x_{\lambda_K+4}[nT_B] := X[(n-\lambda_B)T_B]; x_{\lambda_K+5}[nT_B] := \\ &X[(n-\lambda_B+1)T_B]; \dots x_{\lambda_K+\lambda_B+3}[nT_B] := X[(n-1)T_B] \end{aligned} \tag{24}$$

It also follows that:

$$\begin{aligned} x_{\lambda_K+3}[(n+1)T_B] &= x_{\lambda_K+4}[nT_B]; x_{\lambda_K+4}[(n+1)T_B] \\ &= x_{\lambda_K+5}[nT_B]; \dots x_{\lambda_K+\lambda_B+3}[(n+1)T_B] : \\ &= X[nT_B] \end{aligned} \tag{25}$$

With the above definitions, the overall closed-loop feedback equations in Eq. (23) can thus be written as:

$$\begin{Bmatrix} \mathbf{X_{aug}}[(n+1)T_B] \\ x_{\lambda_K+3}[(n+1)T_B] \\ x_{\lambda_K+4}[(n+1)T_B] \\ \cdot \\ \cdot \\ \cdot \\ x_{\lambda_K+\lambda_B+2}[(n+1)T_B] \\ x_{\lambda_K+\lambda_B+3}[(n+1)T_B] \end{Bmatrix} = \begin{bmatrix} \mathbf{\Phi^r_{aug}} & \mathbf{\Gamma^r_{aug}}\frac{B}{T_B} & -\mathbf{\Gamma^r_{aug}}\frac{B}{T_B} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 1 & 0 & \dots & 0 \\ \mathbf{0} & \mathbf{0} & 0 & 1 & \dots & 0 \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \dots & \cdot \\ \mathbf{0} & \mathbf{0} & 0 & 0 & \dots & 1 \\ \mathbf{1} & \mathbf{0} & 0 & 0 & \dots & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{X_{aug}}[nT_B] \\ x_{\lambda_K+3}[nT_B] \\ x_{\lambda_K+4}[nT_B] \\ \cdot \\ \cdot \\ \cdot \\ x_{\lambda_K+\lambda_B+2}[nT_B] \\ x_{\lambda_K+\lambda_B+3}[nT_B] \end{Bmatrix} \tag{26}$$

Eq. (26) can be simply put as:

$$\begin{cases} \mathbf{X_{cl}}[(n+1)T_B] = \mathbf{\Phi_{cl}}\mathbf{X_{cl}}[nT_B] \\ Y[nT_B] = \mathbf{\Psi_{cl}}\mathbf{X_{cl}}[nT_B] \end{cases} \tag{27}$$

where, $\mathbf{\Psi_{cl}} = [\mathbf{\Psi_{aug}}\ 0\ 0\ \dots\ 0]_{1\times(\lambda_K+\lambda_B+3)}$. These state-space equations represent the dynamics of the overall closed loop time-delayed dual-rate haptics controller in the discrete-time domain. It is quite apparent that the order of this state-space representation is $\lambda_K + \lambda_B + 3$, that can become very high for larger time delays and higher sampling frequencies.

Now, in order to evaluate the stability bounds of the controller, we make use of discrete-time Lyapunov theory. According to the Lyapunov theory, the discrete-time system defined by Eq. (27) is asymptotically stable if and only if there exists a unique positive definite matrix $\mathbf{P}$, for a given positive definite matrix $\mathbf{Q}$, satisfying the following discrete-time Lyapunov equation (Franklin et al. 1998):

$$\mathbf{\Phi_{cl}}\mathbf{P}\mathbf{\Phi_{cl}^T} - \mathbf{P} + \mathbf{Q} = 0 \tag{28}$$

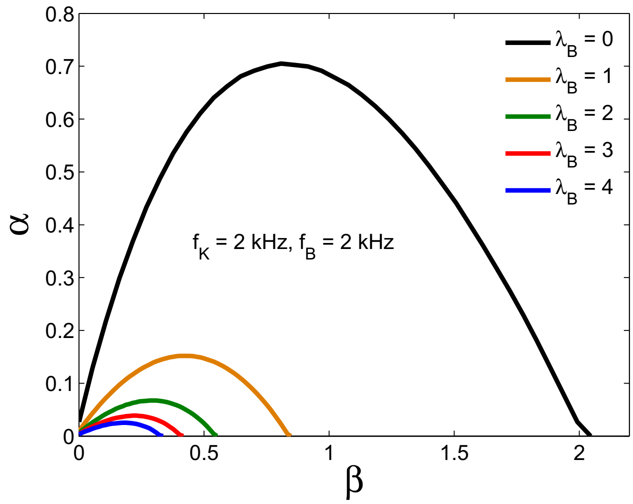
The matrices $\mathbf{P}$ and $\mathbf{Q}$ in this case are obviously of dimensions $(\lambda_K + \lambda_B + 3) \times (\lambda_K + \lambda_B + 3)$. Furthermore, V $= \mathbf{X_{cl}^T}\mathbf{P}\mathbf{X_{cl}}$ is the Lyapunov function for the system and it also follows that, $\Delta V = -\mathbf{X_{cl}^T}\mathbf{Q}\mathbf{X_{cl}}$ is the gradient (Franklin et al. 1998). The matrix $\mathbf{Q}$ will be simply assumed to be an identity matrix, i.e., $\mathbf{Q} = \mathbf{I}_{(\lambda_K+\lambda_B+3)\times(\lambda_K+\lambda_B+3)}$.

Based on Eq. (28), the stability limits of the time-delayed dual-rate haptics controller are determined by gradually increasing the values of the virtual damping B, and for each value of B, evaluating the maximum value of the virtual stiffness K, for which the condition yields a positive definite matrix $\mathbf{P}$. In order to present the results, we use the non-dimensional parameters suggested in Hulin et al. (2014), Mashayekhi et al. (2018). The non-dimensional parameters are presented in Table 1. Using Eq. (28), the stability boundaries for few selected sample period combinations and time delays are shown in Figs. 2 and 3. These stability boundaries have been obtained for a constant value of $T_B = 0.5$ ms, and the value of T_K has been chosen as 0.5, 0.25, 0.1 and 0.05 ms, respectively. These values correspond to values of $N = \{1, 2, 5 \text{ and } 10\}$, respectively. The value of λ has been chosen as 0.5, 1, 1.5 and 2 ms, for the small delay case, and as 10 and 20 ms, for the large time delay case, respectively. The smaller values of time delay λ are more realistic of a virtual wall rendering scenario (Gil et al. 2009; Mashayekhi et al. 2018), while the higher values may sometimes feature in cases such as deformable object rendering (Cavusoglu and Tendick 2000; Barbagli et al. 2002). These values of time delay thus span both the best-case and the worst-case scenarios in which the dual-rate controllers may find their application. It is to be noted that these values of λ correspond to values of $\lambda_B = \{1, 2, 3 \text{ and } 4\}$, for the small delay case, and $\lambda_B = \{10, 20\}$, for the large time delay case, respectively. Furthermore, the stability boundaries have been obtained for $I = 0.000929$ Kg m^2 and $b = 0.0264$ *Nms/rad* (Koul et al. 2017). With regard to the Figs. 2 and 3, it can be seen that the presence of time delay affects the stability boundaries for both the uniform-rate and the dual-rate sampling schemes. In fact, as the time delay increases the stable range of both the virtual stiffness K (non-dimensional α) and virtual damping B (non-dimensional β) decreases. In particular, a delay of $\lambda = 0.5$ ms ($\lambda_B = 1$) leads to a reduction in the maximum achievable virtual stiffness K by 78.5, 82, 86.4 and 88.7%, for $T_B = 0.5$ ms and $N = \{1, 2, 5 \text{ and } 10\}$, respectively. These results suggest that the reduction in the stability limits due to time delay depends on the value of the time delay relative to the sampling periods i.e., λ_B and λ_K, rather than the magnitude of the time delay (λ) itself.

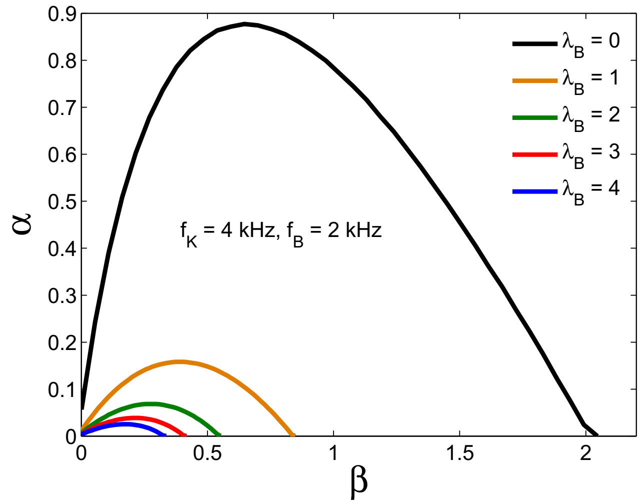
Although, Eqs. (27) and (28) enable us to determine the stability boundaries of the dual-rate haptics controller with delayed feedback, however, the higher model order of such equations makes the computation very demanding, particularly for large time-delays and higher position loop sampling rates. For instance, the stability boundaries in Fig. 3b were developed by solving matrices of order 63 and 123, respectively. This is not feasible in cases where the stability bounds need to be determined in realtime during actual haptic interactions. In Sect. 5, we use a model order reduction technique and seek to obtain lower order models that can yield the same stability limits as given by the higher order models developed in this section.

Table 1 Dimensionless parameters in accordance with Hulin et al. (2014); Mashayekhi et al. (2018)

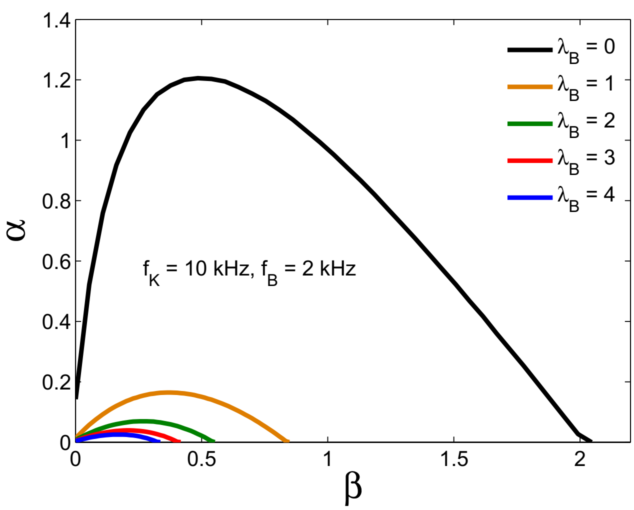
Parameter	Variable	Dimensionless variable
Delay in stiffness loop	λ	$\lambda_K := \frac{\lambda}{T_K}$
Delay in damping loop	λ	$\lambda_B := \frac{\lambda}{T_B}$
Physical damping	b	$\delta := \frac{bT_B}{I}$
Virtual damping	B	$\beta := \frac{BT_B}{I}$
Virtual stiffness	K	$\alpha := \frac{KT_B^2}{I}$



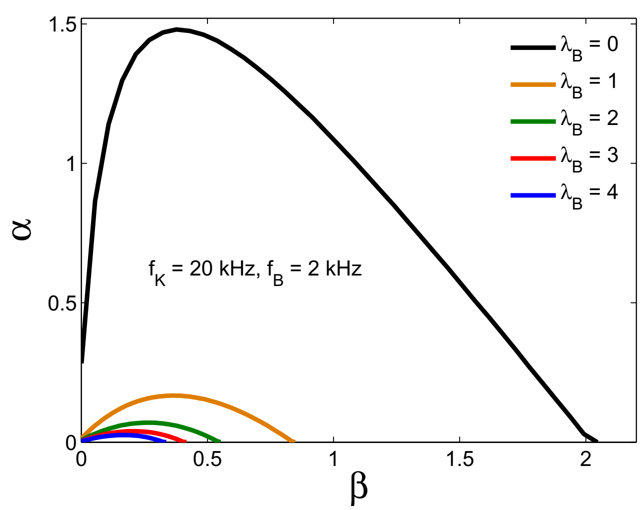
(a) Non-dimensional *Z-width* plots for $f_K = 2$ kHz, $f_B = 2$ kHz (N = 1) and $\lambda_B = \{0, 1, 2, 3, 4\}$



(b) Non-dimensional *Z-width* plots for $f_K = 4$ kHz, $f_B = 2$ kHz (N = 2) and $\lambda_B = \{0, 1, 2, 3, 4\}$



(c) Non-dimensional *Z-width* plots for $f_K = 10$ kHz, $f_B = 2$ kHz (N = 5) and $\lambda_B = \{0, 1, 2, 3, 4\}$

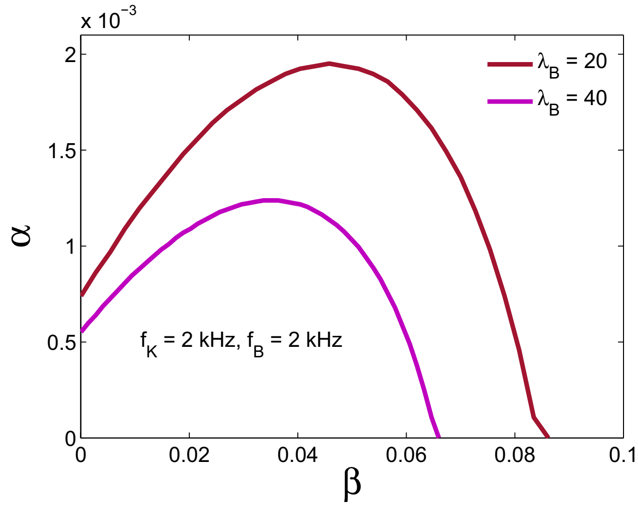


(d) Non-dimensional *Z-width* plots for $f_K = 20$ kHz, $f_B = 2$ kHz (N = 10) and $\lambda_B = \{0, 1, 2, 3, 4\}$

Fig. 2 Stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane for the dual-rate haptics controller for both non-delayed and delayed feedback cases and small values of time delay using the exact discrete-time method. The stability boundaries have been obtained for $T_B = 0.5$ ms ($f_B = 2$ kHz) and different values of f_K corresponding to N = {1, 2, 5, 10}. The values of λ_B correspond to time delays of $\lambda = \{0, 0.5, 1, 1.5, 2\}$ ms

4 Stability analysis using equivalent continuous-time method

In this section, an equivalent continuous-time method is used to determine the stability bounds of the dual-rate haptics controller depicted in Fig. 1. Unlike the exact discrete-time method in which the order of the system depends on the magnitude of the time delay and sampling rate combinations, here no such restriction is observed. The applicability of this method is, however, valid for only low input frequencies, that are relatively smaller than the Nyquist frequency (Gil et al. 2009; Hulin et al. 2008). Since the dual-rate haptics controllers employ high sampling frequencies for virtual



(a) Non-dimensional *Z-width* plots for $f_K = 2$ kHz, $f_B = 2$ kHz (N = 1) and $\lambda_B = \{20, 40\}$

(b) Non-dimensional *Z-width* plots for $f_K = 4$ kHz, $f_B = 2$ kHz (N = 2) and $\lambda_B = \{20, 40\}$

Fig. 3 Stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane for the dual-rate haptics controller for large values of time delay for $T_B = 0.5$ ms ($f_B = 2$ kHz). The values of f_K correspond to N = {1, 2} and the values of λ_B correspond to $\lambda = \{10, 20\}$ ms

wall rendering, this condition is reasonably satisfied, particularly for moderate wall stiffnesses.

In order to represent the dual-rate haptics controller entirely in the continuous-time domain, we approximate the sample and hold operations as a time delay equal to half of the corresponding sampling periods (Gil et al. 2009; Colonnese and Okamura 2016), and the discrete derivative term by the transfer function $se^{-\frac{sT_B}{2}}$ (Mashayekhi et al. 2018, 2019). These delay terms are lumped together with the corresponding feedback time delay terms $z_1^{-\lambda_K}$ and $z_2^{-\lambda_B}$, respectively. It is interesting to note that haptics controllers designed based on this assumption have been found to have improved performance (Gillespie and Cutkosky 1996).

As mentioned in Sect. 2, the continuous-time transfer function of the haptic device (plant) is given by:

$$D(s) = \frac{1}{Is^2 + bs} \tag{29}$$

Due to the presence of the time delay terms in the feedback loops, the transfer function representation of the controller will, in principle, have an infinite order (Franklin et al. 1998). In order to simplify the analysis, and to express the characteristic equation of the controller as a rational function in s, we approximate the combined delay terms in the feedback loops by using first order *Pade approximation*. The first order *Pade approximants* of the combined delay terms in the continuous-time domain are given as Franklin et al. (1998):

$$\begin{cases} e^{-sT_K(\lambda_K+\frac{1}{2})} = \dfrac{1 - \frac{sT_K}{2}(\lambda_K + \frac{1}{2})}{1 + \frac{sT_K}{2}(\lambda_K + \frac{1}{2})} & : \textit{Position feedback loop} \\ e^{-sT_B(\lambda_B+1)} = \dfrac{1 - \frac{sT_B}{2}(\lambda_B + 1)}{1 + \frac{sT_B}{2}(\lambda_B + 1)} & : \textit{Velocity feedback loop} \end{cases} \tag{30}$$

Since, the two feedback loops are in parallel, the combined feedback transfer function can be written as:

$$E(s) = K\left[\frac{1 - \frac{sT_K}{2}(\lambda_K + \frac{1}{2})}{1 + \frac{sT_K}{2}(\lambda_K + \frac{1}{2})}\right] + Bs\left[\frac{1 - \frac{sT_B}{2}(\lambda_B + 1)}{1 + \frac{sT_B}{2}(\lambda_B + 1)}\right] \tag{31}$$

The overall closed-loop transfer function of the controller is given as:

$$G(s) = \frac{D(s)}{1 + D(s)E(s)} \tag{32}$$

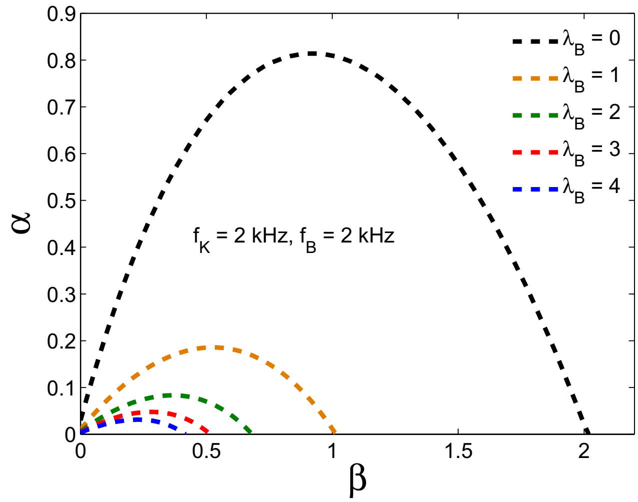
The characteristic equation of the controller is thereby given by:

$$1 + D(s)E(s) = 0 \tag{33}$$

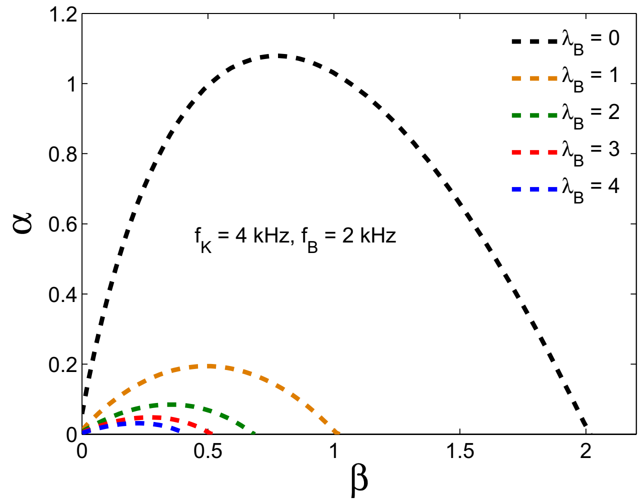
which after substitution leads to:

$$\alpha_c s^4 + \beta_c s^3 + \gamma_c s^2 + \delta_c s + \epsilon_c = 0 \tag{34}$$

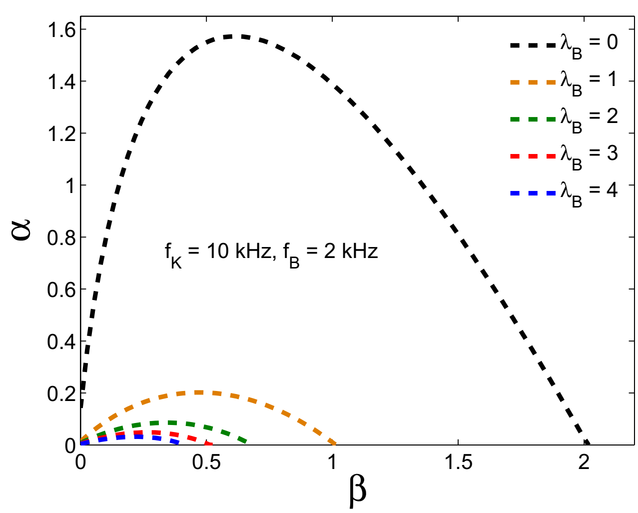
where the coefficients involved in Eq. (34) are defined as:



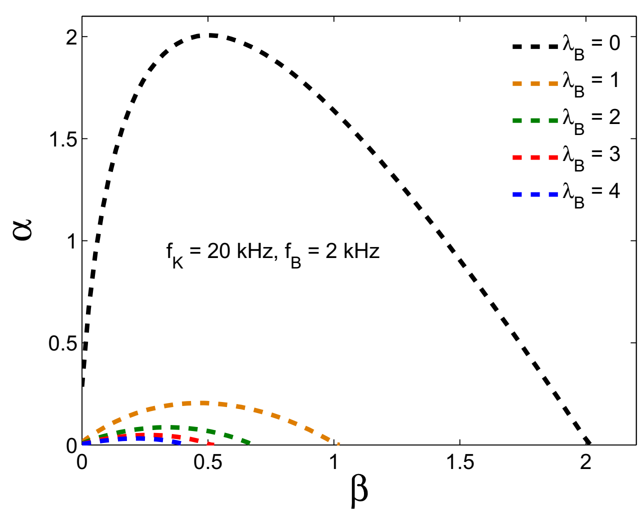
(a) Non-dimensional *Z-width* plots for $f_K = 2$ kHz, $f_B = 2$ kHz (N = 1) and $\lambda_B = \{0, 1, 2, 3, 4\}$



(b) Non-dimensional *Z-width* plots for $f_K = 4$ kHz, $f_B = 2$ kHz (N = 2) and $\lambda_B = \{0, 1, 2, 3, 4\}$



(c) Non-dimensional *Z-width* plots for $f_K = 10$ kHz, $f_B = 2$ kHz (N = 5) and $\lambda_B = \{0, 1, 2, 3, 4\}$



(d) Non-dimensional *Z-width* plots for $f_K = 20$ kHz, $f_B = 2$ kHz (N = 10) and $\lambda_B = \{0, 1, 2, 3, 4\}$

Fig. 4 Stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane for the dual-rate haptics controller for both non-delayed and delayed feedback cases and small values of time delay using the equivalent continuous-time method. The stability boundaries have been obtained for $T_B = 0.5$ ms ($f_B = 2$ kHz) and different values of f_K corresponding to N = $\{1, 2, 5, 10\}$. The values of λ_B correspond to time delays of $\lambda = \{0, 0.5, 1, 1.5, 2\}$ ms

$$\alpha_c = \frac{IT_B^2}{4N}(N\lambda_B + \frac{1}{2})(\lambda_B + 1); \beta_c = \frac{IT_B}{2N}(2N\lambda_B + N + \frac{1}{2}) + \frac{T_B^2}{4N}(b - B)(N\lambda_B + \frac{1}{2})(\lambda_B + 1); \gamma_c = I + \frac{bT_B}{2N}(2N\lambda_B + N + \frac{1}{2}) - \frac{KT_B^2}{4N}(N\lambda_B + \frac{1}{2})(\lambda_B + 1) + \frac{BT_B}{2N}(\frac{1}{2} - N); \delta_c = b + B + \frac{KT_B}{2N}(N - \frac{1}{2}); \epsilon_c = K$$

This method, thus, leads to a 4*th* order characteristic equation, independent of the value of time delays and the sampling rates combinations.

According to the Routh-Hurwitz criterion, for stability we must have:

$$\beta_c > 0; \lambda_c > 0; \frac{\lambda_c \delta_c - \beta_c \epsilon_c}{\lambda_c} > 0 \quad \text{where,} \quad \lambda_c = \frac{\beta_c \gamma_c - \alpha_c \delta_c}{\beta_c} \tag{35}$$

Using the same parameter values as were used in the previous section, the stability boundaries obtained using the equivalent continuous-time method for small time delays are depicted in Fig. 4. As can be seen, the results of both the methods follow the same trend, however, the stability bounds in this case are somewhat higher in value than those obtained using the exact discrete-time method, and thus, are less conservative. This is expected due to the approximate nature of the analysis. In particular, the difference is attributed to the use of the *Pade approximation* for the time delay terms and due to the frequency distortion caused due to the continuous-time representation of the otherwise hybrid haptics controller. Although, here, the results have been provided for small time delays only, but, a similar trend was observed for larger delays also.

5 Reduced order modeling of the time-delayed dual-rate haptics controller

The state-space formulation of time-delayed dual-rate haptics controller presented in Sect. 3 requires the augmentation of the state vector due to the presence of the time delays in the two feedback loops. Such a state augmentation leads to an increase in the system order and consequently makes the stability analysis a computationally intensive task. The augmented states, however, do not represent any physical states and they are merely used to put the state-space equations in standard forms that are amenable for analytical solutions. This intuitively suggests that such augmented states are not as dominant as the other states of the system, and it might be possible to reduce the complexity of the system by somehow reducing its order. The root-loci plots of the controller indeed reinforce this insight as they reveal that despite the order of the system, there are at most three eigenvalues that lie in the vicinty of the unit circle, with which the stability of the controller is associated. The high orders of the controller can be reduced to lower orders by using the model order reduction techniques (Antoulas et al. 2015; Ghosh and Senroy 2013). It is noteworthy to mention, however, that while the higher order models capture the complete dynamics of the system, the reduced order models only approximate the dynamics of the system within certain limitations.

In this section, we employ model order reduction technique based on the balanced truncation method (Ghosh and Senroy 2013; Pernebo and Silverman 1982), to reduce the order of the system in Eq. (27). A balanced truncation method is used as it preserves the stability of the system during the reduction process and can be easily applied to state-space models. A brief introduction of the method is given first, and then, it is applied for the model of the dual-rate haptics controller given by Eq. (27).

Consider a linear time invariant discrete-time system of order p described by the state-space equations of the form:

$$\begin{cases} \mathbf{X}[(k+1)T] = \mathbf{A_d}\mathbf{X}[kT] + \mathbf{B_d}U[kT] \\ \quad Y[kT] = \mathbf{C_d}\mathbf{X}[kT] \end{cases} \tag{36}$$

Broadly speaking, our aim is to obtain a model of order m ($m << p$) which approximates the time response, frequency response and other desirable properties of the above system, but essentially retains its stability. The degree to which this can be accomplished, depends obviously on the choice of the technique being used.

In balanced truncation method, the discrete controllability and discrete observability gramians play an important role in the model reduction process. For the system represented by Eq. (36), the two are given by Franklin et al. (1998):

$$\mathbf{W_{con}} = \sum_{k=0}^{\infty} \mathbf{A_d}^k \mathbf{B_d} \mathbf{B_d}^T (\mathbf{A_d}^T)^k \tag{37}$$

$$\mathbf{W_{obs}} = \sum_{k=0}^{\infty} (\mathbf{A_d}^T)^k \mathbf{C_d}^T \mathbf{C_d} \mathbf{A_d}^k \tag{38}$$

The two gramians can be obtained by solving the following two discrete-time Lyapunov equations (Franklin et al. 1998):

$$\mathbf{A_d}\mathbf{W_{con}}\mathbf{A_d}^T - \mathbf{W_{con}} + \mathbf{B_d}\mathbf{B_d}^T = 0 \tag{39}$$

$$\mathbf{A_d}^T\mathbf{W_{obs}}\mathbf{A_d} - \mathbf{W_{obs}} + \mathbf{C_d}^T\mathbf{C_d} = 0 \tag{40}$$

The importance of $\mathbf{W_{con}}$ and $\mathbf{W_{obs}}$ is that, if they are positive definite, then the system defined by Eq. (36), is controllable as well as observable. In general, $\mathbf{W_{con}}$ and $\mathbf{W_{obs}}$ are not equal and are also basis dependent (Franklin et al. 1998). A linear transformation of Eq. (36) changes its basis and consequently changes both $\mathbf{W_{con}}$ and $\mathbf{W_{obs}}$. It is known (Moore 1981), that there exists a particular basis, and hence, a linear transformation, in which $\mathbf{W_{con}}$ and $\mathbf{W_{obs}}$ are equal and diagonal, i.e.,

$$\mathbf{W_{con}} = \mathbf{W_{obs}} = \mathbf{diag}\{\sigma_1, \sigma_2, ...\sigma_p\} \tag{41}$$

The state-space representation of the system (Eq. (36)) in such a basis is referred to as balanced realization. The $\sigma's$ are the eigenvalues of the product of the controllability and observability gramians. The square roots of $\sigma's$ are referred to as the Hankel Singular Values (HSVs), and they represent the energy associated with the states of the system. A higher values of HSV signifies that the corresponding state has a high energy and dominates the system dynamics. The balanced realization and the balanced truncation method of model order reduction retains the states of the higher order model that have higher HSVs, and discard the ones with lower HSVs. The usefulness of a balanced realization of a system is that every state is equally controllable and observable simultaneously.

Now, in order to obtain the balanced realization of the system represented by Eq. (36), a similarity transformation can be used by applying a linear coordinate transformation of the form $\mathbf{X} = \mathbf{T}_s\hat{\mathbf{X}}$ on the state vector $\mathbf{X}$, such that the system (Eq. 36) transforms to:

$$\begin{cases} \hat{\mathbf{X}}[(n+1)T] = \hat{\mathbf{A}}_\mathbf{d}\hat{\mathbf{X}}[nT] + \hat{\mathbf{B}}_\mathbf{d}U[nT] \\ Y[nT] = \hat{\mathbf{C}}_\mathbf{d}\hat{\mathbf{X}}[nT] \end{cases} \tag{42}$$

where, the transformed matrices are given by Ghosh and Senroy (2013):

$$\hat{\mathbf{A}}_\mathbf{d} = \mathbf{T}_\mathbf{s}^{-1}\mathbf{A}_\mathbf{d}\mathbf{T}_\mathbf{s};\ \hat{\mathbf{B}}_\mathbf{d} = \mathbf{T}_\mathbf{s}^{-1}\mathbf{B}_\mathbf{d};\quad \text{and} \quad \hat{\mathbf{C}}_\mathbf{d} = \mathbf{C}_\mathbf{d}\mathbf{T}_\mathbf{s} \tag{43}$$

The controllability and the observability gramians of the transformed system are related to the original system as Ghosh and Senroy (2013):

$$\hat{\mathbf{W}}_\mathbf{con} = \mathbf{T}_\mathbf{s}^{-1}\mathbf{W}_\mathbf{con}(\mathbf{T}_\mathbf{s}^{T})^{-1} \quad \text{and} \quad \hat{\mathbf{W}}_\mathbf{obs} = \mathbf{T}_\mathbf{s}^{T}\mathbf{W}_\mathbf{obs}\mathbf{T}_\mathbf{s} \tag{44}$$

As mentioned earlier, in balanced realization form, the controllability and the observability gramians are equal and diagonal, i.e.,

$$\hat{\mathbf{W}}_\mathbf{con} = \hat{\mathbf{W}}_\mathbf{obs} = \hat{\mathbf{W}} = \mathbf{diag}\{\sigma_1, \sigma_2, \dots\sigma_p\} \tag{45}$$

The algorithm proposed in Laub et al. (1987), Pernebo and Silverman (1982) can be used to determine the transformation matrix $\mathbf{T}_\mathbf{s}$. First, the Cholesky factor $\mathbf{L}_\mathbf{c}$ (a lower triangular matrix) is evaluated by the Cholesky decomposition of $\mathbf{W}_\mathbf{con}$, i.e., by solving $\mathbf{W}_\mathbf{con} = \mathbf{L}_\mathbf{c}\mathbf{L}_\mathbf{c}^{T}$. Afterwards, the singular value decomposition of the matrix product $\mathbf{L}_\mathbf{c}^{T}\mathbf{W}_\mathbf{obs}\mathbf{L}_\mathbf{c}$ is obtained. This involves decomposing the matrix product into the form $\mathbf{L}_\mathbf{c}^{T}\mathbf{W}_\mathbf{obs}\mathbf{L}_\mathbf{c} = \mathbf{U}\hat{\mathbf{W}}^2\mathbf{U}^T$. Once $\mathbf{L}_\mathbf{c}$, $\mathbf{U}$ and $\hat{\mathbf{W}}$ are obtained, the transformation matrix $\mathbf{T}_\mathbf{s}$ can be determined by:

$$\mathbf{T}_\mathbf{s} = \mathbf{L}_\mathbf{c}\mathbf{U}\sqrt{\hat{\mathbf{W}}} \tag{46}$$

Since, $\hat{\mathbf{W}}$ is a diagonal matrix, $\sqrt{\hat{\mathbf{W}}} = \mathbf{diag}\{\sqrt{\sigma_1}, \sqrt{\sigma_2}, \dots\sqrt{\sigma_p}\}$.

Next, the HSVs can be evaluated using:

$$HSV_i = \sqrt{eig_i(\mathbf{W}_\mathbf{con}\mathbf{W}_\mathbf{obs})} \quad i = 1, 2, 3 \dots p \tag{47}$$

where, HSV_i and eig_i denote the i th Hankel Singular Value and the eigenvalue, respectively.

Based on the HSVs, the states with higher values are retained in the reduced model, while those with lower values are discarded. For instance, if the first m HSVs are far greater than the rest, then the original system represented by Eq. (36) can be reduced to order m without significantly affecting the input-output dynamics of the original system.

The balanced truncation based reduced order model of Eq. (36) is then obtained by partitioning the diagonalized gramian $\hat{\mathbf{W}}$ as:

$$\hat{\mathbf{W}} = \begin{bmatrix} \hat{\mathbf{W}}_\mathbf{1} & 0 \\ 0 & \hat{\mathbf{W}}_\mathbf{2} \end{bmatrix} \tag{48}$$

where,

$$\hat{\mathbf{W}}_\mathbf{1} = \mathbf{diag}\{\sigma_1, \sigma_2 \dots \sigma_m\} \quad \text{and} \quad \hat{\mathbf{W}}_\mathbf{2} = \mathbf{diag}\{\sigma_{m+1}, \sigma_{m+2} \dots \sigma_p\} \tag{49}$$

In a similar manner, the system matrices involved in Eq. (42), are also partitioned as:

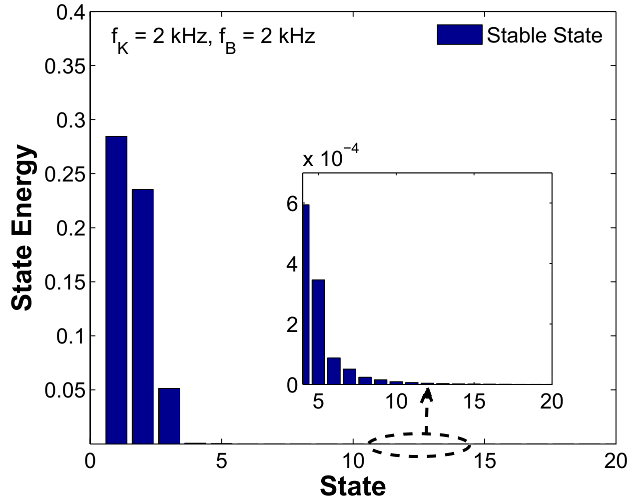
$$\hat{\mathbf{A}}_\mathbf{d} = \begin{bmatrix} \hat{\mathbf{A}}_\mathbf{d11} & \hat{\mathbf{A}}_\mathbf{d12} \\ \hat{\mathbf{A}}_\mathbf{d21} & \hat{\mathbf{A}}_\mathbf{d22} \end{bmatrix};\hat{\mathbf{B}}_\mathbf{d} = \left\{\hat{\mathbf{B}}_\mathbf{d1}\ \hat{\mathbf{B}}_\mathbf{d2}\right\}^T \quad \text{and} \quad \hat{\mathbf{C}}_\mathbf{d} = \left\{\hat{\mathbf{C}}_\mathbf{d1}\ \hat{\mathbf{C}}_\mathbf{d2}\right\} \tag{50}$$

Based on the knowledge of these partitioned matrices, the reduced order model (order m) of the system can be expressed as:

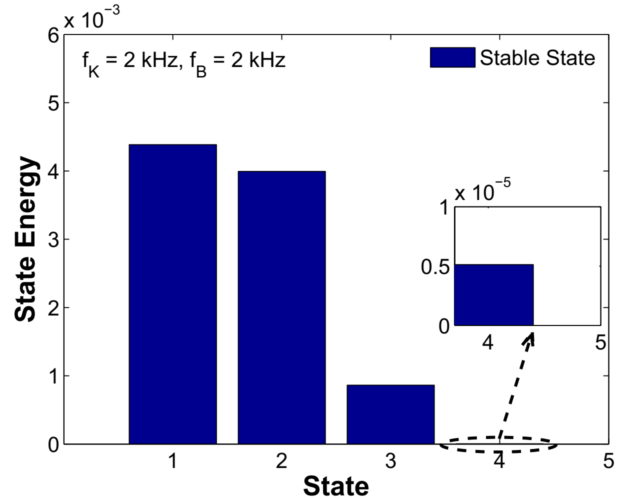
$$\begin{cases} \hat{\mathbf{X}}_\mathbf{r}[(n+1)T] = \hat{\mathbf{A}}_\mathbf{d11}\hat{\mathbf{X}}_\mathbf{r}[nT] + \hat{\mathbf{B}}_\mathbf{d1}U[nT] \\ Y[nT] = \hat{\mathbf{C}}_\mathbf{d1}\hat{\mathbf{X}}_\mathbf{r}[nT] \end{cases} \tag{51}$$

This reduced order model is an approximation to the original higher order model in Eq. (36), however, both the models have the same stability characteristics (Antoulas et al. 2015).

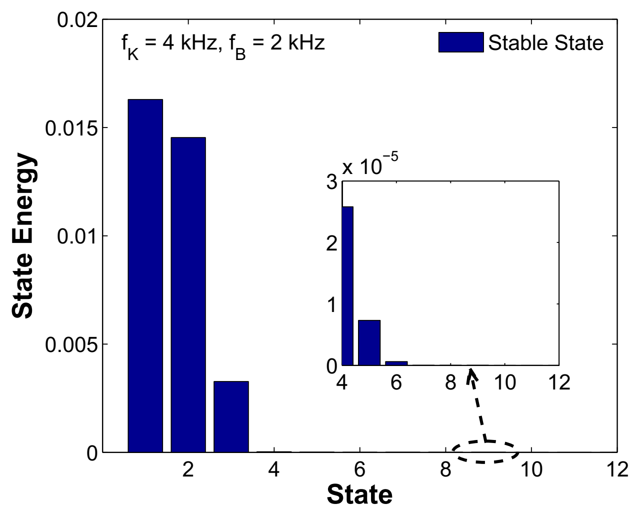
We applied this model order reduction method to the closed loop state-space model of the time-delayed dual-rate haptics controller given by Eq. (27). Many sampling rate combinations and time delays, representative of the practical dual-rate haptic rendering scenario were chosen (stable as well as unstable) in order to determine the likely order of the reduced order model. In all the cases, only the first three HSVs were found to be dominant, whereas, the others were found to be negligible. The HSVs for a few selected cases are depicted in Fig. 5, and as can be seen, whether the system is stable or unstable, only the first three system states contribute mainly towards the system energy. It is to be noted that the notion of state energy, as depicted in Fig. 5, is different from system energy. The former refers to the energy of a particular state variable while as the latter indicates the energy of the system as a whole. Similar results were obtained for several other parameter combinations but they are not provided here due to space constraints. This empirical result leads to the conclusion that for the most likely values of time delay and sample rate combinations, a



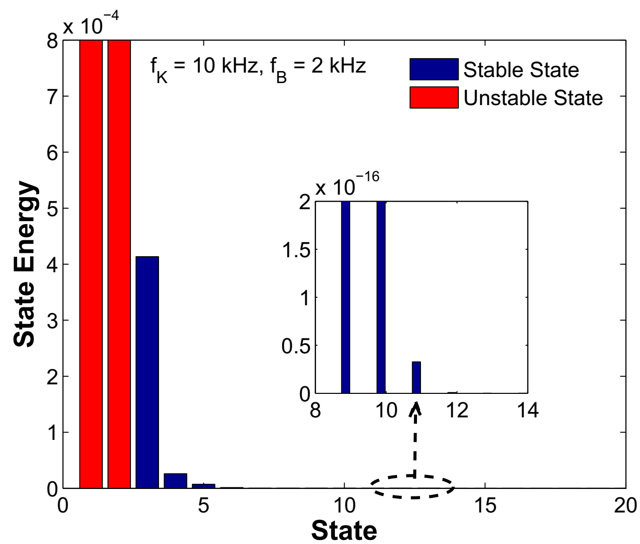
(a) Hankel singular values (state energy) of the system states for f_K = 2 kHz, f_B = 2 kHz (N = 1), λ_B = 20, B = 0.1 Nms/rad and K = 5 Nm/rad



(b) Hankel singular values (state energy) of the system states for f_K = 2 kHz, f_B = 2 kHz (N = 1), λ_B = 1, B = 1 Nms/rad and K = 400 Nm/rad



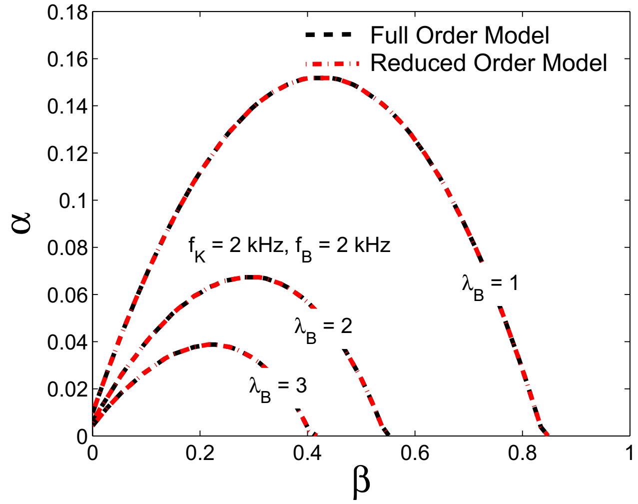
(c) Hankel singular values (state energy) of the system states for f_K = 4 kHz, f_B = 2 kHz (N = 2), λ_B = 3, B = 0.5 Nms/rad and K = 100 Nm/rad



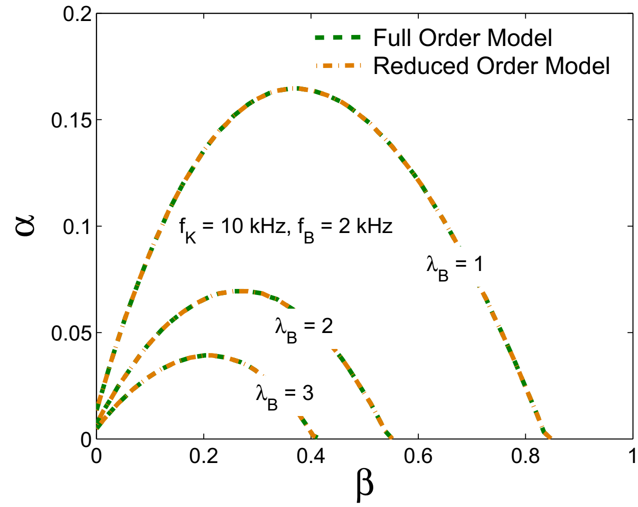
(d) Hankel singular values (state energy) of the system states for f_K = 10 kHz, f_B = 2 kHz (N = 5), λ_B = 4, B = 0.2 Nms/rad and K = 120 Nm/rad

Fig. 5 The contribution of system states towards the system energy (Hankel singular values) for various sample-rate combinations and time delays. Sub-figures **a** and **b** are for uniform-rate sampling scheme, **c** and **d** are for dual-rate sampling schemes, respectively. For all the cases only the first three system states dominate the system energy and consequently determine the stability of the system

time-delayed dual-rate haptics controller can be represented by a third order reduced model for the purpose of stability analysis. Using the third order reduced models, the stability analysis of the dual-rate haptics controller was carried out using the discrete-time Lyapunov theory. The stability boundaries obtained for the frequency combinations $f_K = 2\text{kHz}, f_B = 2\text{kHz}$ and $f_K = 10\text{kHz}, f_B = 2\text{kHz}$ and for smaller values of time delay are shown in Fig. 6, and for larger time delays are depicted in Fig. 7. As expected, the stability boundaries obtained based on the third order reduced models match exactly the ones obtained earlier using the higher order models. This finding is in fact valid



(a) Stability boundaries (Non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 2kHz, f_B = 2kHz$ (N = 1) and for $\lambda_B = \{1, 2, 3, 4\}$



(b) Stability boundaries (Non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 10kHz, f_B = 2kHz$ (N = 5) and for $\lambda_B = \{1, 2, 3, 4\}$

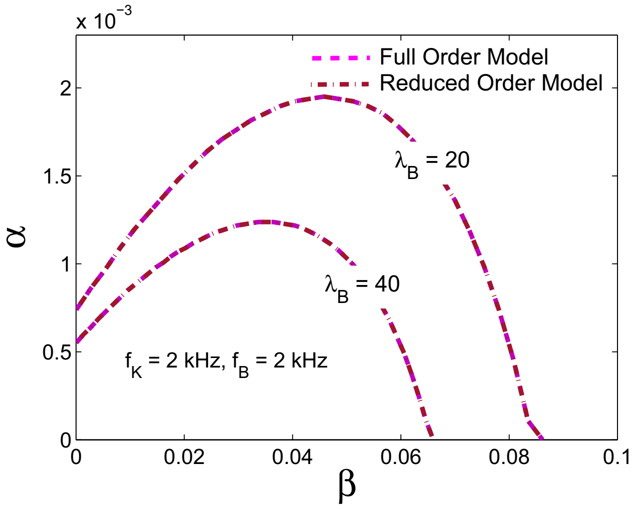
Fig. 6 A comparison of the stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane obtained using the full order models (exact discrete-time method) and third order reduced models (balanced truncation method) for small values of time delay. Subfigure **a** and **b** correspond to uniform-rate and dual-rate sampling schemes, respectively. **a** Stability boundaries (non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 2kHz, f_B = 2kHz$ (N = 1) and for $\lambda_B = \{20, 40\}$. **b** Stability boundaries (non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 4kHz, f_B = 2kHz$ (N = 2) and for $\lambda_B = \{20, 40\}$

for both the uniform-rate as well dual-rate time-delayed haptics controllers.

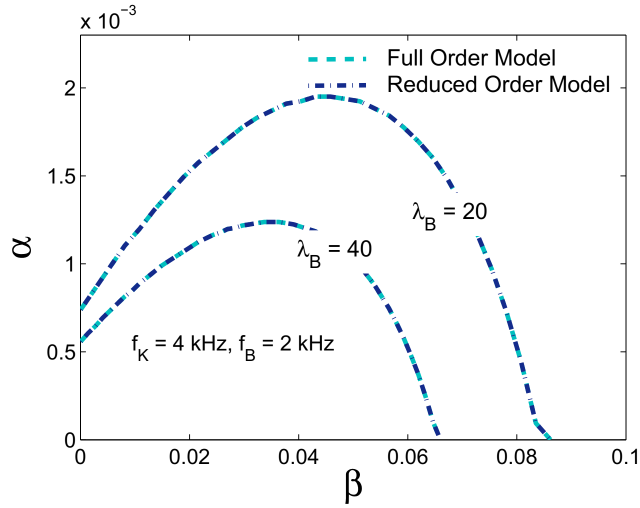
6 Discussions

The analysis carried out in the previous sections reveals that time delay affects both the stable ranges of virtual damping B (non-dimensional β) and the virtual stiffness K (non-dimensional α). The results presented in Figs. 2, 3 and 4 highlight that for a given sampling rate combination, an increase in time delay leads to a reduction in the stability bounds for both the uniform-rate and dual-rate sampling schemes. These results are in agreement with the results available in the literature for uniform-rate sampling scheme. As a matter of fact, the results obtained here using the exact discrete-time method, match exactly with the results of Gil et al. (2004) for non-delayed uniform-rate haptics controllers, and with the results of Hulin et al. (2014) and Mashayekhi et al. (2018) for time-delayed uniform-rate haptics controllers. The comparison is depicted in Fig. 8. The current work, is thus more general in the sense that it is not only applicable for time-delayed dual-rate haptics controllers, but is equally valid for uniform-rate haptics controllers, whether delayed or non-delayed.

In order to augment the theoretical results presented in the previous sections, several simulations were carried out to determine the stability limits of the time-delayed dual-rate haptics controller. For the same purpose, a Simulink model was developed in MATLAB® in accordance with Fig. 1. The device parameters that were used earlier in the theoretical analysis were also used for the Simulink model. For a particular combination of T_B, N and λ, the simulations were carried out by gradually increasing the value of B, and for each value, determining the maximum value of K for which the simulation response remained stable. For a particular value of B, the values of K were incremented in fixed steps of 5 *Nm/rad*, and for each value, the simulation response was observed. The value of K was increased, in case the response turned out to be stable, otherwise, the value of K was noted and the same represented the maximum renderable stiffness for those particular parameter combinations. The values of B were incremented in fixed steps of 0.05 *Nms/rad*, and the procedure was repeated until the entire stability boundary was generated. Since the simulations were time consuming, the results were obtained for only two frequency combinations: one representing the uniform-rate sampling scheme



(a) Stability boundaries (Non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 2kHz, f_B = 2kHz$ (N = 1) and for $\lambda_B = \{20, 40\}$



(b) Stability boundaries (Non-dimensional *Z-width* plots) obtained using the full order and reduced order models for the frequency combination $f_K = 4kHz, f_B = 2kHz$ (N = 2) and for $\lambda_B = \{20, 40\}$

Fig. 7 A comparison of the stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane obtained using the full order models (exact discrete-time method) and third order reduced models (balanced truncation method) for large values of time delay. Sub-figure **a** and **b** correspond to uniform-rate and dual-rate sampling schemes, respectively

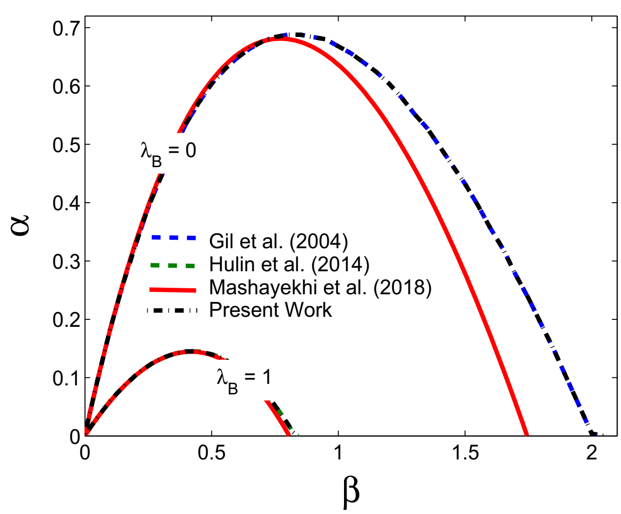
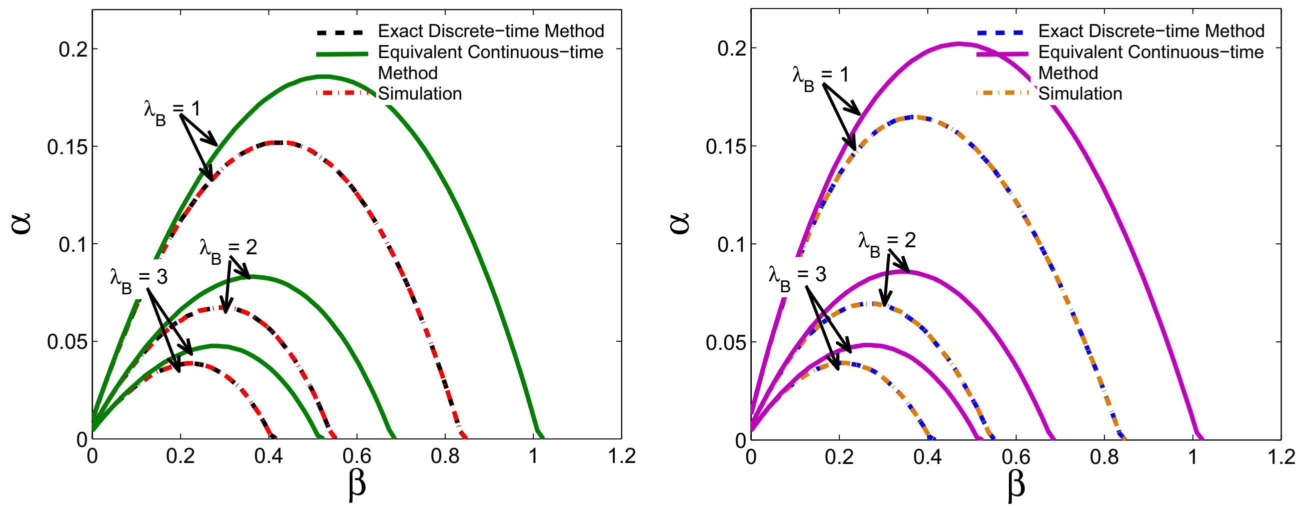


Fig. 8 A comparison of the stability boundaries (non-dimensional *Z-width* plots) obtained using the exact discrete-time method with several existing methods in the literature for uniform-rate haptic rendering without delay ($\lambda_B = 0$) and for a one sample delay ($\lambda_B = 1$). The device parameters used are $I = 0.5\ Kgm^2$, $b = 0.1$ Nms/rad and $T = 0.001\ s$

(f_K = 2 kHz, f_B = 2 kHz) and the other representing the dual-rate sampling scheme (f_K = 10 kHz, f_B = 2 kHz). Moreover, the results were only obtained for smaller time delays ($\lambda = \{0, 0.5, 1, 1.5, 2\}$ ms). The simulation results are shown in Fig. 9 in terms of the non-dimensional parameters α and β along with the theoretical results obtained in the previous sections. The simulation results can be seen to overlap the results obtained using the exact-discrete time method for both the uniform-rate as well as dual-rate sampling schemes regardless of the values of time delay and sample rate combinations, and they are more conservative than the results obtained using the equivalent continuous-time method. This suggests that Simulink uses the same exact transformation sequences as were used in the exact discrete-time method. A close-up of Fig. 9, depicted in Fig. 10, however, indicates that both the theoretical methods and the simulations give the same stability bounds for small values of virtual damping B. Thus, in applications, where only small damping is to be emulated, any of the two methods can be used to predict the stability of the controller.

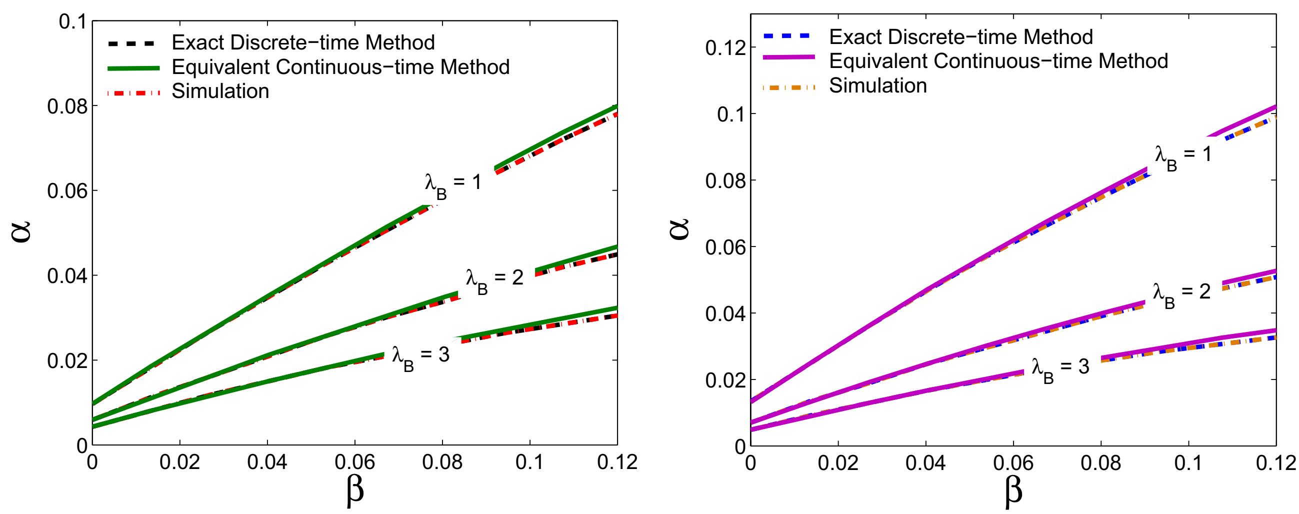
To gain deeper insights into the combined effect of time delay and dual-rate sampling on the stability of haptics controllers, several stability boundaries have been obtained for different sample rate combinations and constant values of time delay. The stability boundaries are depicted in Fig. 11. Figure 11a shows the stability boundaries for low sampling frequency and high time delay (λ = 10 ms), while as, Fig. 11b depicts the stability boundaries for high sampling frequency and lower time delay (λ = 1 ms). It can



(a) Simulation results (Non-dimensional *Z-width* plots) for $f_K = 2$ kHz, $f_B = 2$ kHz (N = 1) and $\lambda_B = \{1, 2, 3, 4\}$

(b) Simulation results (Non-dimensional *Z-width* plots) for $f_K = 10$ kHz, $f_B = 2$ kHz (N = 5) and $\lambda_B = \{1, 2, 3, 4\}$

Fig. 9 A comparison of the stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane obtained using the two theoretical methods and the simulations for small values of time-delay $\lambda = [0.5, 1, 1.5, 2]$ ms. Sub-figures **a** and **b** correspond to uniform-rate haptic rendering and dual-rate haptic rendering, respectively

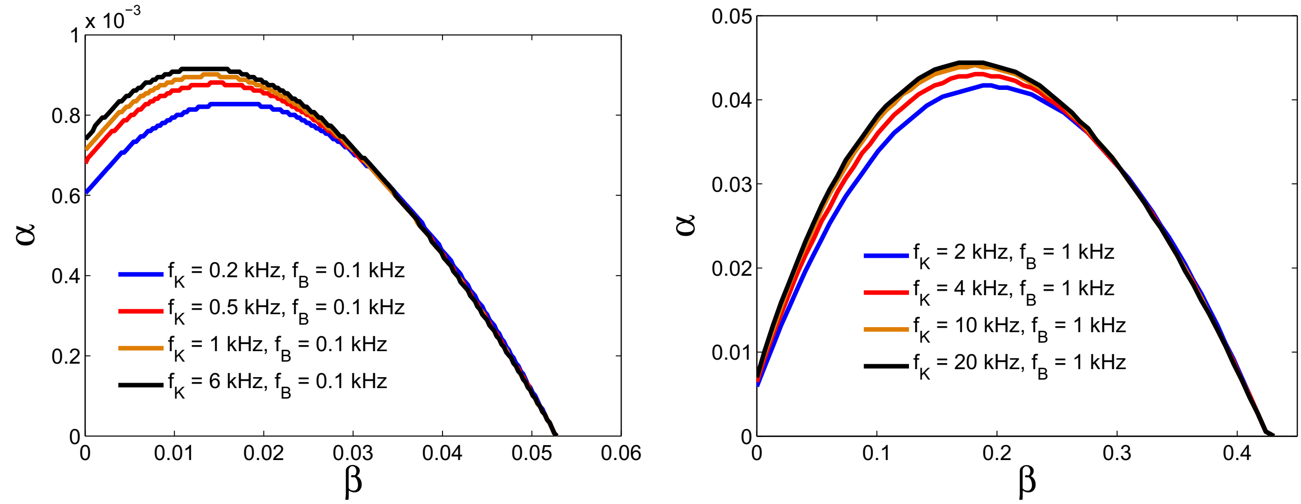


(a) A close-up of the stability boundaries (Non-dimensional *Z-width* plots) for $f_K = 2$ kHz, $f_B = 2$ kHz (N = 1), $\lambda_B = \{1, 2, 3, 4\}$ and small values of virtual damping B

(b) A close-up of the stability boundaries (Non-dimensional *Z-width* plots) for $f_K = 10$ kHz, $f_B = 2$ kHz (N = 5), $\lambda_B = \{1, 2, 3, 4\}$ and small values of virtual damping B

Fig. 10 A comparison of the stability boundaries (non-dimensional *Z-width* plots) in the $\alpha - \beta$ plane obtained using the two theoretical methods and the simulations for small values of virtual damping B, and small time delays $\lambda = [0.5, 1, 1.5, 2]$ ms. Sub-figures **a** and **b** correspond to uniform-rate haptic rendering and dual-rate haptic rendering, respectively

be seen that for a constant time delay, increasing the sampling frequency of the position loop while maintaining that of the velocity loop constant, enhances the stable range of virtual stiffness K while the stable range of virtual damping B remains constant. The constant value of virtual damping B is obvious as the sampling frequency of the velocity loops



(a) Stability boundaries (Non-dimensional *Z-width* plots) for dual-rate haptic rendering at different sampling frequency combinations and a constant time delay of 10 ms

(b) Stability boundaries (Non-dimensional *Z-width* plots) for dual-rate haptic rendering at different sampling frequency combinations and a constant time delay of 1 ms

Fig. 11 Combined effect of dual-rate sampling scheme and time delay on the stability boundaries (non-dimensional *Z-width* plots) in dual-rate haptic rendering

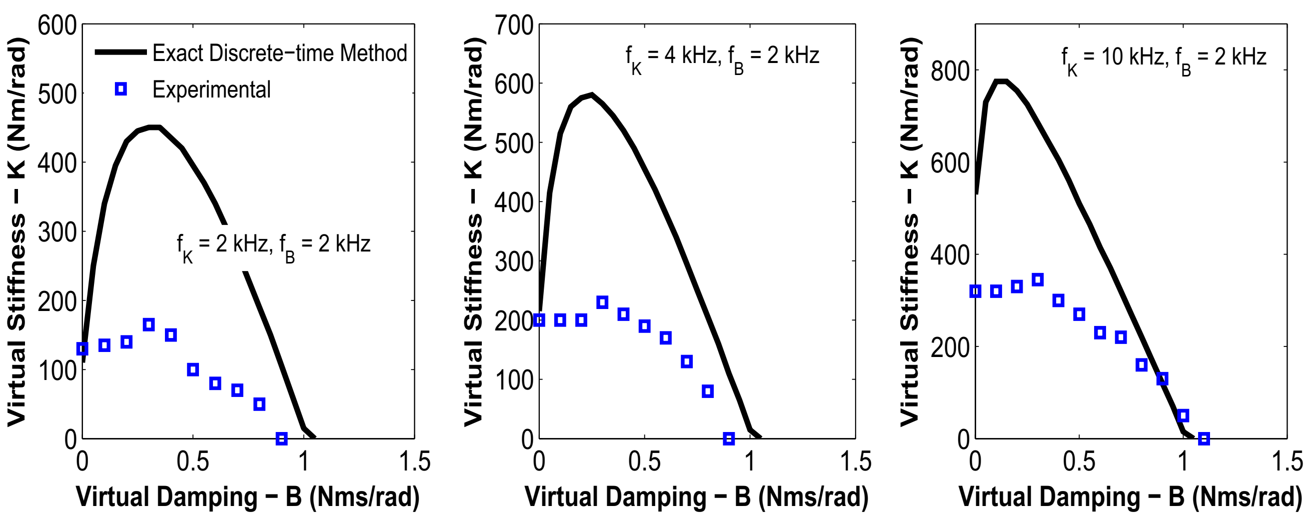


Fig. 12 A comparison of the results obtained using the exact discrete-time method with the experimental results reported in Koul et al. (2017), for different sampling frequency combinations, and for $\lambda_B = 2$, $\lambda_K = 0$

is constant in all the cases considered. However, it is to be noted that higher values of f_B increase the stable range of B. Furthermore, it is also evident that as the sampling frequency of the position loop f_K increases, the relative increment in the maximum virtual stiffness K diminishes. The reason for this behavior is that for a constant λ, an increase in the value of f_K corresponds to an increase in the value of λ_K, which in turn implies that the effect of time delay outweighs the effect of higher sampling frequencies and consequently leads to a very marginal improvement in the performance. The same observation has previously been made by Abbott and Okamura (2005) and Hulin et al. (2014) for uniform-rate haptics controllers.

A comparison of the results obtained using the exact discrete-time method (Sect. 3) has also been made with the experimental results obtained by Koul et al. (2017). The experiments reported in Koul et al. (2017) have been conducted on a 1-DoF haptic device utilizing a Maxon RE25

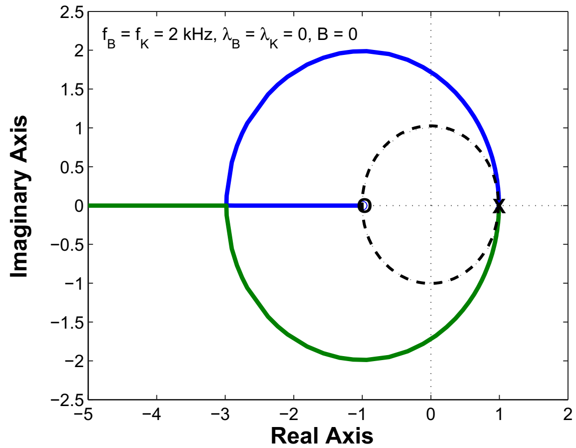
DC motor as an actuator with an integrated 3 channel 1000 CPR encoder. A virtual wall is implemented using a field programmable gate array (FPGA) by employing the dual-rate sampling scheme. While the processing delay in the FPGA is negligible, a time delay of 1 ms exists in the velocity loop implementation due to low pass filtering, for a sampling frequency $f_B = 2$ kHz and cutoff frequency $f_{cut} = 200$ Hz. The results of the comparison are depicted in Fig. 12. It can be seen that both the theoretical and the experimental results yield the same value of the maximum stable virtual damping B, however, an appreciable difference exists in the range of the stable virtual stiffness K. This difference can be attributed to the non-linear phenomena like sensor quantization, coulomb friction, actuator saturation, unilateral constraint etc. that creep in during the actual haptic interaction (Diolaiti et al. 2006; Abbott and Okamura 2005). A simulation based approach reported in Chawda et al. (2015), Koul et al. (2017) indeed shows that the incorporation of non-linearities like sensor quantization and Coulomb friction in the linear model of the haptics controller leads to a drastic reduction in the theoretical stability boundaries. Besides this, the mismatch between the theoretical and the experimental results is due to the simplified device model (rigid model) that the author's have used in the theoretical analysis. For instance, we have not taken the flexibility of the device into consideration in the theoretical analysis which is otherwise known to affect the stability boundaries in haptic rendering (Díaz and Gil 2010). As the main focus of the present work is to demonstrate how dual-rate haptics controllers with time delayed feedback can be mathematically modeled and to assess the possibility of reducing the high order models using the model order reduction framework, therefore it was preferred to make use of such a simplified device model. Another reason for the mismatch is that the theoretical stability limits correspond to the onset of asymptomatic instability while as the ones reported experimentally correspond to the onset of self-sustained oscillations i.e., the limit cycles (Koul et al. 2017; Hayward et al. 1997). Nevertheless, the experimental stability boundaries are a subset of the theoretical stability boundaries and as such they can be used in the prediction of the stability limits.

The use of model order reduction in Sect. 5 has facilitated the stability analysis of the dual-rate haptics controller and reduced the computational time for the estimation of the stability bounds. In Table 2, the computational time for the assessment of stability of the dual-rate controller for certain selected control parameters and virtual wall impedances is provided. It can be seen that for all the iterations, the reduced order model takes lesser time than the full order model for stability assessment. While the time difference seems to be small for a single virtual wall parameter combination, however, for the determination of the entire stability boundaries this difference is much significant.

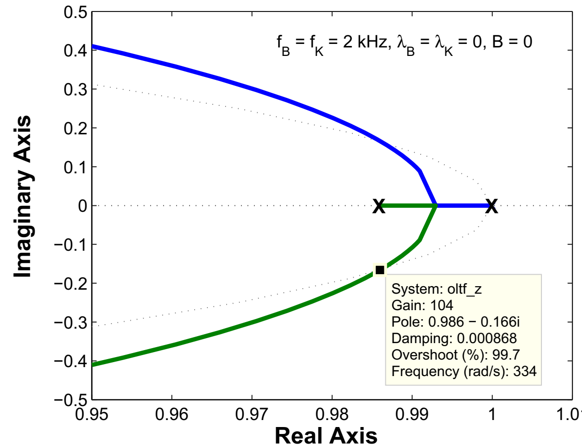
For the sampling rate combinations and time delays considered in this paper, it has been found that a third order reduced model is sufficient for assessing the stability of the dual-rate haptics controller irrespective of the order of the full order models. In order to gain insights as to why third order models can successfully predict the stability of the dual-rate haptics controller we make use of the discrete-time root-locus technique in this section. We first consider the case of non-delayed haptic rendering and subsequently analyze the delayed case. Figure 13a shows the discrete-time root-locus plot for non-delayed ($\lambda_B = \lambda_K = 0$) haptic rendering of virtual stiffness ($B = 0$) for $f_B = f_K = 2$ kHz. For these parameter combinations, the controller has two real open loop poles (z = 1 and z = 0.9859) and one real open loop zero (z = –0.9953). Since the two poles are very close to each other, only one **X** symbol is visible. These two open loop poles, on which the stability of the controller depends, come from the discretized second order plant transfer function $D(s) = \frac{1}{Is^2+bs}$. As the open loop poles lie either in the interior or on the unit circle boundary, the controller starts from being stable at $K = 0$. However, owing to the presence of a zero at infinity, the stability is maintained only upto a certain range of K (critical stiffness), and for values beyond that the controller is unstable. From the magnified view of the root-locus plot in Fig. 13b, it can be seen that as the value of K increases, the two real open loop poles move towards each other, subsequently transform into complex conjugates and eventually at the critical stiffness of $K = 104$ Nm/rad they cross the unit circle boundary, and accordingly lead to the onset of

Table 2 A comparison of the computational time incurred for stability assessment at selected control and virtual wall parameters, using the full order discrete-time models, and the third order reduced models

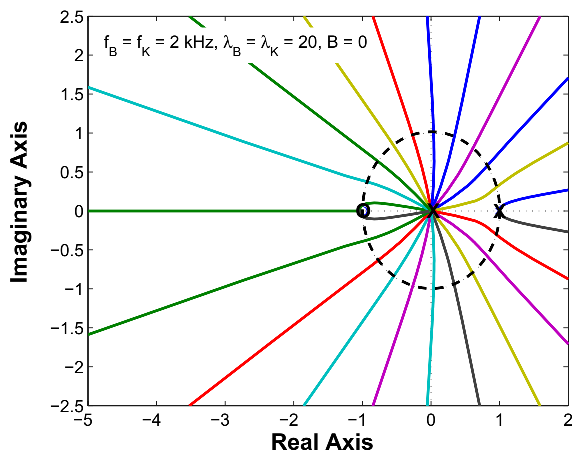
Control parameters	Virtual wall parameters	Computational time	
		Full order model	Reduced order model
$f_B = 2$ kHz, $f_K = 4$ kHz, $\lambda_B = 20$, $\lambda_K = 40$	B = 0.1 Nms/rad, K = 6 Nm/rad	0.111648 s	0.031608 s
$f_B = 2$ kHz, $f_K = 10$ kHz, $\lambda_B = 4$, $\lambda_K = 20$	B = 0.5 Nms/rad, K = 50 Nm/rad	0.053761 s	0.029079 s
$f_B = 2$ kHz, $f_K = 20$ kHz, $\lambda_B = 3$, $\lambda_K = 30$	B = 0.2 Nms/rad, K = 100 Nm/rad	0.065634 s	0.030635 s



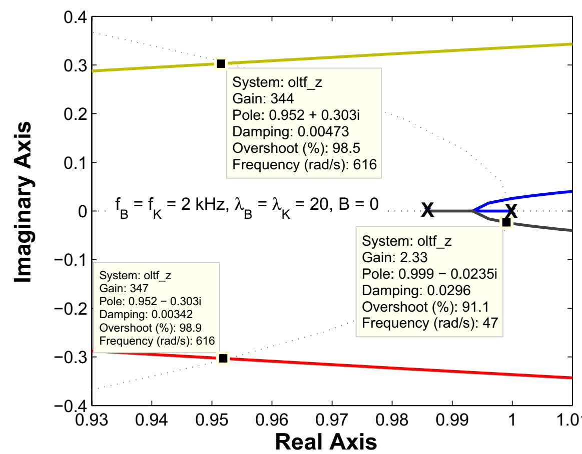
(a) Root-locus for $f_B = f_K = 2$ kHz (N = 1), $\lambda_B = \lambda_K = 0$, and $B = 0$ Nms/rad



(b) Magnified view of the root-locus shown in (a) depicting the onset of instability and the critical feedback gain



(c) Root-locus for $f_B = f_K = 2$ kHz (N = 1), $\lambda_B = \lambda_K = 20$, and $B = 0$ Nms/rad



(d) Magnified view of the root-locus shown in (c) depicting the onset of instability and the critical feedback gain

Fig. 13 Root-loci for few selected values of f_B, f_K, λ_B, λ_K, and B with K as feedback gain

controller instability. The initial location of these poles that depends on the device and control parameters determines the range of stable virtual stiffness K and the boundary crossing frequency (334 rad/s). For instance, higher values of device damping b and sampling frequencies f_B and f_K shift the poles towards the origin and consequently lead to an increase in the critical stiffness. This finding is well known from earlier works in the haptics literature (Colgate and Brown 1994; Gil et al. 2004; Minsky et al. 1990). For the delayed case, we consider $\lambda_B = \lambda_K = 20$, while keeping the other control parameters same as used for the non-delayed case. Figure 13c depicts the root-locus plot for such a case and its magnified view is shown in Fig. 13d. It is interesting to note that for the delayed case, the initial location (for $K = 0$) of the open loop poles (z = 1 and z = 0.9859) is invariant and does not depend on the magnitude of the time delay. The presence of time delay, however, introduces poles of multiplicity 20 at the origin of the unit circle (z = 0). As was the case with the non-delayed rendering, the controller is stable initially as all the open loop poles lie either in the interior or on the unit circle boundary. The controller, however, loses stability after a certain gain value (critical stiffness) as all the poles except one cross the unit circle boundary. While for the non-delayed case, the two complex conjugate poles crossed the unit circle boundary at the same gain value of $K = 104$ Nm/rad and accordingly set the stability limit. However, for the delayed case, all the open loop poles do not cross the unit circle boundary at the same gain value and accordingly the stability limit depends on the poles that cross the unit circle first and mark the onset of instability. From the magnified view of the root-locus plot depicted in Fig. 13d, it can be easily seen that it is the open loop poles (z = 1 and z = 0.9859), associated with the plant transfer function, that happen to cross the unit circle first at the critical stiffness of $K = 2.33$ Nm/rad, while the poles associated with the time delay cross at higher gain values. Two of such time delay poles cross the unit circle at higher gains of $K = 344$ Nm/rad, and $K = 347$ Nm/rad, respectively, as depicted in Fig. 13d. The only effect of time delay is that it reduces the unit circle boundary crossing frequency from 344 rad/s for the non-delayed case to 47 rad/s for the delayed case, respectively. As a consequence of this the critical stiffness, reduces from $K = 104$ Nm/rad for the non-delayed case to $K = 2.33$ Nm/rad for the delayed case, respectively. It is important to note, however, that whether the controller involves non-delayed or delayed feedback, the stability limits happen to be associated with the poles that belong to the device transfer function, which in this case is of second order. This explains as to why the model order reduction technique yields the stability limits from a reduced second order controller models for virtual stiffness rendering ($B = 0$), and from reduced third order models for the more general virtual wall rendering cases where B is non-zero. For the general case of the virtual wall rendering, the controller model also involves a pole due to the discrete differentiation of the position signal, and accordingly the controller model is of third order.

Finally, it is important to reiterate that while the application of model order reduction in Sect. 5 has considerably simplified the determination of the stability limits of the dual-rate controller, however, the reduced order models only approximate the higher order models, and thus, they should not be used as their replacement for other purposes (for instance, in system identification) without prior analysis. The reduced order models based on balanced truncation may not preserve the time-response, frequency response, static gain, structure (for e.g., block structure of matrices) of the higher order models but they preserve the stability and passivity, which were of primary concern in this paper. However, there exist several hybrid MOR techniques that can be used to preserve many combinations of properties like stability and static gain etc. (Antoulas et al. 2015).

7 Conclusions and future work

In this paper, the stability of time-delayed dual-rate haptics controller is studied using two distinct theoretical methods and numerical simulations. The main objective is to establish the effect of the combined presence of time delay and dual-rate sampling on the stability bounds of such controllers. The results of the stability analysis reveal that time delay significantly reduces the stability region of the dual-rate haptics controller. For instance, a time delay of 0.5 ms is found to reduce the maximum renderable virtual stiffness by 78.5, 82, 86.4 and 88.7% when the velocity loop is sampled at 2 kHz and the position loop is sampled at variable sample rates of 2, 4, 10 and 20 kHz, respectively. Furthermore, the analysis reveals that the stability bounds of the dual-rate haptics controller depend strongly on the value of time delay relative to the sample rates of the controller. For lower values, an increase in the controller sampling rates leads to an appreciable improvement in performance, while as for higher values, a marginal improvement is observed. The determination of the stability bounds is shown to be facilitated by employing a model order reduction technique based on the balanced truncation method. For typical control parameters, it is found that the dual-rate haptics controller can be approximated by a third order model for the purpose of stability analysis. A theoretical proof of this result, and its extension to multiple DoF haptic systems will be the focus of our future work.

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Declarations

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